

What Carries Deep-Noise Robustness in a Mamba-3 Block? Input-Independent Scans and Noise-Dependent Gating under Cross-Condition Shift

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Abstract

The robustness of Mamba-style selective state-space models is often attributed to input-dependent selectivity. We test this with controlled ablations of one Mamba-3 block on bearing vibration data under cross-condition shift (XJTU-SY, one speed-and-load pair run in both directions, training and evaluation bearings disjoint), with matched-noise training and final-epoch macro-F1 under a fixed budget. All central contrasts were run with training and evaluation noise drawn independently. Making the scan’s dynamics and read/write coefficients input-independent does not remove the block’s advantage over a time-invariant S4D: on the original transition the fully frozen block leads by +21.3, +23.3, +20.3 pp at 0, −2 and −6 dB, and on the reverse transition by +27.5, +26.7, +21.1 pp. Removing the multiplicative output gate costs +5.4, +9.3, +32.6 pp in the fully frozen block, resolved only at −6 dB; only in this host and direction does the cost grow significantly with noise depth, and without its gate the block still exceeds S4D at 0 and −2 dB but not at −6 dB. Multiplicative gating exceeds an additive branch of identical size by +23.3 pp at −6 dB. On Paderborn bearings the gate-removal cost replicates when training and evaluation share bearings. Grafting a gate onto S4D, per-class results and Paderborn with disjoint bearings were run only under an earlier noise protocol and are descriptive. The evidence concerns one block, one shift pair per dataset and five seeds.

Keywords: state-space models, Mamba, input selectivity, gating,

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1. Introduction

Selective state-space models (SSMs) of the Mamba family have rapidly become default sequence backbones, and a common narrative has formed around *why* they work: input-dependent selectivity—the ability to modulate state dynamics per token—is credited with filtering noise, adapting to distribution shift, and outperforming time-invariant predecessors such as S4D [1]. [2, 3] This belief motivates a growing family of “selective” architectures, yet recent evidence questions it: a systematic comparison on 59 clean multivariate time-series classification benchmarks (MONSTER and UEA) finds the *simpler, time-invariant* S4D consistently ahead of Mamba-family variants in accuracy and efficiency [4], and theoretical analyses identify expressivity regimes where Mamba is provably weaker [5]. Which account is right—and, more importantly, *what exactly should be credited* when a Mamba-style block does prove robust? This paper addresses the second question for one block and one testbed, with a controlled ablation chain whose decision thresholds were fixed before each grid was run. The testbed is bearing vibration diagnosis under operating-condition drift and matched noise, in which the robustness gap is large (up to 18 pp), the evaluation labels are identical across all cells of each transition (hash-verified), and the block can be dissected.

Our approach is deliberately *attributional* rather than *methodological*: we propose no new architecture and claim no state-of-the-art. We take a regime in which a Mamba-style block shows a large, stable robustness advantage over a time-invariant SSM, and ask which ingredient earns it, in three steps with adjudication thresholds fixed before each grid was run:

1. **Freeze the scan.** The input-dependent parts of Δ , A , B , C are replaced by learned constants; in a second arm the trapezoidal weight and rotation angles are as well, so that the scan’s dynamics and read/write coefficients no longer depend on the input. The value stream and the gate input remain input-dependent (Section 3.2).
2. **Remove the gate.** On the frozen base the multiplicative SiLU gate is removed, with the other components and the budget unchanged.
3. **Replace or transplant the gate.** In place, the gate is replaced by an additive branch of identical size and initial weights; across hosts, it is grafted onto plain S4D with an additive control.

Steps 1 and 2 and the in-place replacement were tested with training and evaluation noise drawn independently (Section 4); the graft onto S4D, the per-class analysis, the stem and A ablations and the disjoint-bearing Paderborn setting were run only under an earlier protocol in which the two noise streams were shared by window index, and are reported in Appendix D as descriptive evidence (Table 2). The outcome is a conditional pattern. Making the scan input-independent does not remove the block’s advantage over S4D, on either transition (+21.3, +23.3, +20.3 pp and +27.5, +26.7, +21.1 pp for the fully frozen block). Removing the gate costs +9.4, +15.7, +27.0 pp in the partially frozen block on the original transition, resolved only at -6 dB, and 19.3, 22.7, 22.0 pp on the reverse transition, resolved at every level. Only in the fully frozen block on the original transition does the gate-removal cost grow significantly with noise depth: without its gate that block still exceeds S4D at 0 and -2 dB but not at -6 dB. An additive branch of identical size does not recover the gate’s effect at -6 dB in either host, although the pre-specified rescue test is not assessable. On Paderborn bearings the gate-removal cost replicates when training and evaluation share bearings.

Contributions. (i) A layered freezing experiment on a Mamba-3 block, down to a scan with input-independent dynamics and read/write coefficients, in which the advantage over S4D survives on both transitions of one dataset under independent training and evaluation noise; the frozen coefficients are verified input-independent at the kernel interface. (ii) Same-host tests of the output gate (removal and in-place additive replacement) in a partially and a fully frozen block, together with a descriptive analysis of how the gate’s contribution changes with noise depth. (iii) A coverage table that states, for every operation, whether it was run under independent noise, and a correction of the original noise protocol with a pre-specified re-run. (iv) A released pre-specification and review trail, including the incidents in which automated adjudication departed from the pre-specified criteria.

2. Background and Related Work

2.1. Time-invariant and selective SSMs

Diagonal structured SSMs (S4D) [6] evolve a linear state with input-independent parameters (A , B , C , Δ), acting as a bank of learned linear filters. Mamba-style blocks [1] make Δ , B , C functions of the input token

(selectivity), wrap the scan in a block containing a short causal convolutional stem, a multiplicative SiLU gate on a parallel branch, and residual/normalisation structure; the concrete block used throughout this paper follows the Mamba-3 formulation [7]. The narrative crediting selectivity for robustness draws on theory showing input-dependent dynamics can represent discontinuity-sensitive projections and counteract memory decay [8], and on [9]. We treat these as *hypotheses to be tested*, not established attribution-s: the theoretical results concern representability, which does not by itself determine which component earns empirical robustness in a trained block.

2.2. The unresolved comparison

Empirically the picture is contested. A systematic comparison on 59 clean multivariate time-series classification benchmarks (MONSTER and UEA) finds S4D-based models superior to Mamba variants [4]; that study does not include additive noise or a shift of operating condition between training and evaluation, which is the regime examined here; theoretical work identifies long-range regimes where Mamba is weaker than S4D [5]. Meanwhile, application literatures—including machine-condition monitoring—contain many Mamba-hybrid methods reporting noise robustness [10, 11, 12, 13, 14], but none, to our knowledge, isolates *which component* of the block is responsible under controlled conditions. Our clean-data results agree with the S4D-superiority finding (S4D matches or slightly exceeds the Mamba-style block on clean data). Whether the disagreement in the literature reflects a regime difference is not settled by a single testbed; what we can report is that, in the drift $\times$ noise regime studied here, the Mamba-style block’s advantage over S4D is not removed by freezing the classical selectivity terms.

2.3. Testbed rationale

We use bearing vibration diagnosis under cross-condition drift (speed and load both change) as the testbed, for three properties that matter to the attribution question rather than to the application: (a) the distribution shift is physically generated (a change of rotation speed and radial load between recorded conditions), not simulated; (b) fault signatures are sparse transients—precisely the discontinuous structures selectivity theory predicts selective SSMs should excel at [8], which makes it a plausible arena for the selectivity hypothesis; and (c) the robustness gap between the Mamba-style

block and S4D is large (up to 20 pp), giving the attribution chain high signal. The testbed is two-class (details in Section 3.1); the scope of all claims is bounded accordingly (Section 6).

3. Experimental Protocol

3.1. Testbed, arms, and budget

Vibration windows from the XJTU-SY run-to-failure dataset [15] are split by operating condition: models train on one operating condition (37.5 Hz, 11 kN) and evaluate on another (40 Hz, 10 kN), so speed and load both change (cross-condition), with additive white Gaussian noise applied at matched SNR to training and evaluation ($\{\text{clean}, +10, +6, 0, -2, -6, -10 \text{ dB}\}$). The task is binary fault classification (inner-race vs. outer-race dominant classes; class imbalance makes IR the hard class, Appendix D.10). All arms share an identical training budget (50 epochs, 5 seeds; the per-class grid of Appendix D.10 used 20 epochs), identical splits, and identical evaluation labels, verified by a hash of the label vector: every cell on the original transition carries `eval_sha = 6c20b367...` (Appendix A), recomputed independently during adjudication; the reverse transition and the two Paderborn settings carry their own label hashes (`93424834...`, `53d565d9...` disjoint, `addc1938...` shared). The hash certifies the label vector, not the input windows. In total the study comprises 865 cells across five pre-specified grids of the original design (baseline grid 475; frozen-selectivity 105; component ablation 175; graft 35; per-class 75), run under the earlier noise protocol, and follow-up grids add 645 cells: 210 under independent training and evaluation noise (Sections 4.1–4.3) and 435 under the earlier protocol (Appendix D); Table 2 gives the coverage. Training and evaluation bearings are disjoint in each direction: Bearing2_1, 2_2, 2_4, 2_5 at 37.5 Hz/11 kN and Bearing3_1, 3_3, 3_4, 3_5 at 40 Hz/10 kN; the reverse transition swaps their roles.

Composition of the data. Training uses the bearings of the 37.5 Hz/11 kN condition whose failure mode is inner race or outer race (Bearing2_1 is labelled IR; Bearing2_2, 2_4 and 2_5 are labelled OR); evaluation uses those of the 40 Hz/10 kN condition (Bearing3_3 and 3_4 are IR; Bearing3_1 and 3_5 are OR). Bearings with cage or mixed failure modes are excluded. Windows are non-overlapping segments of 2048 samples of the horizontal channel taken from the fault phase, which is delimited by a kurtosis/RMS onset criterion with a last-15% fallback; the evaluation set holds 3,504 OR and 1,744 IR windows, the training set 4,624 OR and 624 IR windows. Two consequences

140 matter for interpretation. The IR class is learned from a single training bear-
 141 ing, so class-level results may reflect bearing-specific generalisation. And the
 142 five seeds resample initialisation, batch order and noise but not bearings:
 143 there are four independent bearings on each side, and none of the intervals
 144 reported here captures bearing-level uncertainty. One follow-up grid (Sec-
 145 tion 4.2) reverses the transition ($40\text{ Hz}/10\text{ kN} \rightarrow 37.5\text{ Hz}/11\text{ kN}$). It uses the
 146 same eight bearings with their roles swapped, so it tests the other direction
 147 rather than independent bearings; its evaluation set holds 4,624 OR and
 148 624 IR windows, all IR windows coming from Bearing2_1, and its label hash
 149 is 93424834... The 35 Hz/12 kN condition has no inner-race bearing and
 150 cannot form a binary pair.

151 The principal arms are summarised in Table 1.

152 3.2. What the frozen arm freezes, and what the checks verify

153 In the Mamba-3 layer used here, one input projection emits the gate and
 154 value streams, the scan inputs B and C , the step-size and decay terms (Δ
 155 and A , through input-dependent additive terms), a trapezoidal weight and
 156 rotation angles. The frozen arm removes the input-dependent parts of Δ ,
 157 A , B and C and replaces them with learned constants broadcast over time
 158 (for Δ the static bias of the original layer is kept; for A a new constant pa-
 159 rameter plays the role of a static log-decay, passed through the layer’s own
 160 heavy-tailed clamp). The removed projections are deleted, so no parameter
 161 or gradient can leak. **The trapezoidal weight and the rotation angles**
 162 **remain functions of the current input**, as do the stem and the output
 163 gate. The frozen arm is therefore not a time-invariant scan: in the Mamba-3
 164 formulation [7] the input-dependent rotation and trapezoidal weighting also
 165 enter the state update. Appendix D.1 is accordingly a statement about the
 166 classical selectivity terms Δ , A , B , C only. Three further arms close that gap
 167 by the same device: *frozen-trap* replaces the trapezoidal weight by a learned
 168 per-head constant, *frozen-rot* replaces the rotation angles by learned con-
 169 stants (so the rotation becomes a learned position-only phase rather than a
 170 function of the current input), and *frozen-all* does both, so that the scan’s
 171 dynamics and read/write coefficients (Δ , A , B , C , trapezoidal weight, rota-
 172 tion angles) no longer depend on the input. The value stream x and the gate
 173 input z are still computed from the input, as they must be for the layer to
 174 carry any signal; the rotation becomes a learned position-dependent phase.
 175 In each case the corresponding `in_proj` output slice is deleted. Appendix
 176 D.2 reports them. For these arms the construction check adds the decisive

Table 1: Principal experimental arms and parameter counts.

Arm	Description	Params
BM3	Mamba-style block stack (conv stem $k=7 \rightarrow$ selective scan $\rightarrow$ SiLU gate $\rightarrow$ projection), with and without an auxiliary physics loss (Section 3.4)	177,938
BM3-frozen	BM3 with the input-dependent parts of Δ , A , B , C replaced by learned constants; trapezoidal weight, rotation angles, stem, gate and the remaining components unchanged (Section 3.2)	112,410
S4D	Parameter-matched diagonal time-invariant SSM (LayerNorm; no BatchNorm anywhere in either SSM arm)	100,162
S4D-wide	S4D widened to $1.75\times$ parameters (capacity control)	178,498
frozen-conv	Single-component removal on the frozen base: conv stem (Appendix D.3)	112,026
frozen-gate	Single-component removal on the frozen base: SiLU gate (Appendix D.3)	79,642
frozen-A	Single-component removal on the frozen base: heavy-tailed A parameterisation (Appendix D.3)	112,410
frozen-trap	Frozen base with the trapezoidal weight also made a learned constant (Section 3.2)	111,906
frozen-rot	Frozen base with the rotation angles also made learned constants	104,346
frozen-all	Frozen base with both also constant: scan dynamics and read/write coefficients input-independent (value stream and gate still input-dependent)	103,842
frozen-all-gate	frozen-all with the gate rows removed from the input projection (Section 4.1)	71,074
frozen-all+add	frozen-all with $y \odot \text{SiLU}(z)$ replaced by $y + \text{SiLU}(z)$; same parameters and initial weights	103,842
S4D+gate	Plain S4D plus the SiLU gate branch only (graft, Appendix D.4)	116,802
S4D+gate (wm)	Graft rebuilt in the donor’s shape: per layer x, z projected to width 128, S4D at width 128 (state 64, same SSM budget as the host), $W_{\text{out}}(y \odot \text{SiLU}(z))$ (Appendix D.5)	198,978
S4D+branch	Same modules, parameters and initial weights, combined additively: $W_{\text{out}}(y + \text{SiLU}(z))$ (control)	198,978
frozen+add	BM3-frozen with the gate replaced in place: $W_{\text{out}}(y + \text{SiLU}(z))$ instead of $W_{\text{out}}(y \odot \text{SiLU}(z))$;	112,410

177 probe: the tensors handed to the scan are captured for two different inputs
 178 through the same module, and a tensor the arm claims to have frozen is
 179 required to be bitwise identical across them while a tensor left selective is
 180 required to differ (all three arms pass). No kernel modification is needed,
 181 because the scan interface accepts these tensors regardless of their origin.

182 Single-variable discipline was checked by key-level state-dictionary diffs
 183 between two models built under the same seed. For the frozen arm the
 184 diff verifies that every non-target tensor has the same name and shape as in
 185 BM3; for the component-ablation and graft arms it additionally compares the
 186 values of non-target tensors at initialisation. Every differing key is mapped
 187 to an entry in an architecture map. We do not claim byte-level identity of
 188 non-target tensors in the frozen arm, and trained weights naturally differ
 189 between arms; the checks concern how the arms were constructed.

190 3.3. Reporting rule

191 Every cell is trained for a fixed budget of 50 epochs with a cosine learning-
 192 rate schedule that anneals to zero, and the macro-F1 reported in the main
 193 text is that of the *final epoch*, a rule that uses no evaluation data to choose
 194 a model. The original grids recorded instead the maximum over epochs of
 195 macro-F1 on the evaluation split (the harness defines no separate validation
 196 split), and the thresholds pre-specified for those grids were written against
 197 that rule. The arms of the original transition were therefore re-run on the
 198 adjudication band (0, -2, -6 dB) with the original training code, the same
 199 evaluation fingerprint and per-epoch evaluation (final-epoch re-run, earlier
 200 noise protocol; [Appendix D.9](#)). This final-epoch re-run and the independent-
 201 noise re-run of Section 4 are different sets of cells with different coverage
 202 (Table 2). Best-epoch values are a sensitivity analysis. Levels outside the
 203 band (clean, +10, +6, -10 dB) and the per-class grid were not re-run; values
 204 there are best-epoch and auxiliary. Because the final epoch of the same
 205 evaluation condition has been inspected repeatedly during this study, final-
 206 epoch values are not a fresh confirmatory test either.

207 3.4. Confound elimination

208 Before component attribution, three candidate confounds were examined
 209 on the same grid. **Capacity (earlier protocol):** S4D-wide ($1.75\times$ param-
 210 eters) tracks S4D (S4D-wide - S4D: -0.4, -1.0, +0.4 pp at 0, -2, -6 dB
 211 under final-epoch reporting, all intervals including zero), while BM3-frozen
 212 exceeds S4D-wide by +15.9, +20.8, +16.0 pp with intervals excluding zero at

213 all three levels, so this particular increase in S4D’s parameter count does not
 214 explain the gap; the control does not address the change in channel mixing
 215 that gate removal also entails. **Normalisation:** both SSM arms use Layer-
 216 Norm and contain no BatchNorm; the arm that degrades under noise is the
 217 one *without* any batch statistics, which argues against BatchNorm statistics
 218 mismatch as the explanation. Other normalisation inside the Mamba-3 layer
 219 was not varied. **Noise streams:** in the original harness the noise added to
 220 window i was drawn from a generator keyed on (seed, i), and training and e-
 221 valuation used the same seed, so window i of the training set and window i of
 222 the evaluation set carried the same base noise vector. Per-index label agree-
 223 ment between the two sets is 0.549 on XJTU-SY, below the 0.628 expected
 224 by chance, which excludes only the simplest route (memorising the training
 225 noise attached to each index); on Paderborn it is 0.999 (disjoint) and 1.000
 226 (shared). All central contrasts were therefore re-run with the stream keyed on
 227 (seed, split, i), which makes training and evaluation noise independent while
 228 all arms with a given seed share one evaluation noise bank (Section 4.1); the
 229 reverse transition was re-run in the same way (Section 4.2). The component
 230 ablations of the stem and the state parameterisation, the grafts, the additive
 231 substitution on the reverse transition, the per-class grid and the disjoint-
 232 bearing Paderborn setting were not re-run; they carry the earlier protocol in
 233 their captions and are reported in [Appendix D](#). In the arms that were re-run
 234 the means moved by at most 4 pp and no contrast changed direction; this
 235 is an observation about those arms, not a guarantee for the ones that were
 236 not re-run, and the earlier-protocol results are weighted accordingly. **Auxil-
 237 iary physics loss:** an exploratory kinematics-alignment regulariser carried
 238 by the original model contributes -0.1 pp pointwise (curves coincide); all
 239 attribution arms are therefore built on the pure cross-entropy variant.

240 3.5. Pre-specification and adjudication

241 Every adjudicated question in this study had numeric decision thresholds
 242 written to a task card and to disk before the corresponding grid was launched.
 243 This is internal pre-specification, not external registration: the supporting
 244 evidence is local file modification times and run-directory names (released,
 245 [Appendix A](#)), which are not tamper-evident and are not preserved by a
 246 repository checkout, so the trail lets readers inspect the order of events but
 247 cannot prove it. The exception is the follow-up grids from the width-matched
 248 graft onwards (Sections 4.1–4.3 and [Appendix D.5–Appendix D.8](#)): their
 249 thresholds, and for some of them the title decision that depends on their

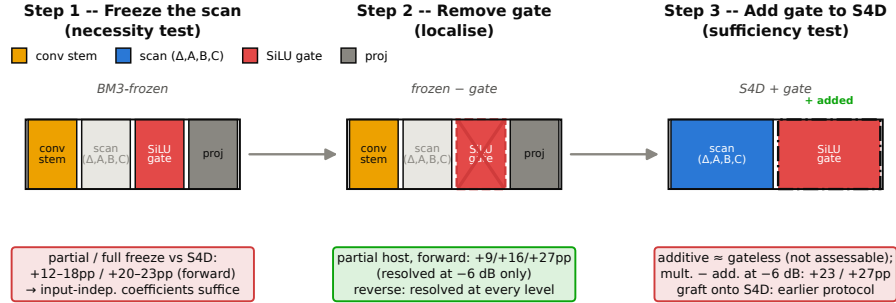
outcome, were committed to the public repository before the first cell was run (including the smoke check), which gives those grids an external time-stamp. Later questions (component ablation, graft) were specified after earlier grids had been seen, as is natural in a staged design. Adjudication was mechanical: an analysis pass applies the thresholds and emits a decision artifact; an independent reviewer pass re-computes aggregates, verifies fingerprints row-by-row, and must cite file paths and modification times for every check. The pre-specified thresholds were binding in practice: two attempted post-hoc criterion substitutions and one stale-artifact false verification were caught and re-adjudicated under the original rules — this is the one place where pre-specification actually bit. [Appendix C](#) details the incidents, the automated executor/reviewer implementation, and the human adjudication gates; all scientific decisions — thresholds, rulings, and interpretations — were made by the authors and are timestamped. In the grids of the original design the adjudication criteria were effect-size thresholds, and paired intervals and tests were reported but not used as criteria. In the later grids (Section 4) the labels *holds*, *weakened* and *does not hold* are defined by whether the paired 95% interval excludes zero at two, one or none of three levels, so interval position is part of the adjudication there. A single repeat of one 2-epoch configuration in a separate process differed by ~ 0.7 pp macro-F1 (non-deterministic kernel reductions; [Appendix B](#)); this is an observed repeat difference, not an estimated variability bound, and arm-level contrasts are judged against seed-level variability (2–9 pp).

4. Results

Throughout, “deep noise” denotes the pre-specified adjudication band $\{0, -2, -6 \text{ dB}\}$; -10 dB is reported only in the earlier-protocol appendix and excluded from adjudication (the time-invariant baseline is near chance there). All values are macro-F1 (%), mean over 5 seeds, at the final epoch of the 50-epoch budget (Section 3.3). **The main text reports only results obtained with independent training and evaluation noise** (Sections 4.1–4.4). Results obtained while training and evaluation noise were shared by index (Section 3.4) are collected in [Appendix D](#) and summarised, with their evidence level, in Section 4.5. Table 2 shows which operation was run under which protocol on which host and transition. Figure 1 summarises the chain.

Table 2: Coverage of the study by protocol. I: run with independent training and evaluation noise (main text). E: run under the earlier protocol ([Appendix D](#)). -: not run. P: partially frozen host (BM3-frozen); F: fully frozen host. “Freeze” is the contrast with S4D (BM3-frozen – S4D for P, frozen–all – S4D for F); “gate removal” and “additive substitution” are applied inside the host. Graft, stem/ A removal and per-class recall have no fully frozen counterpart.

Operation	XJTU forward		XJTU reverse		PU shared		PU disjoint	
	P	F	P	F	P	F	P	F
Freeze (vs S4D)	I	I	I	I	I	–	E	E
Gate removal	I	I	I	–	I	–	E	–
Additive substitution	I	I	E	–	–	–	E	–
Graft onto S4D (with additive control)	E	–	E	–	–	–	–	–
Stem and A removal	E	–	–	–	–	–	–	–
Per-class recall	E	–	–	–	–	–	–	–



Conditional on this block, testbed and budget: input-independent scan coefficients suffice; gate cost grows with noise only in the forward fully frozen block.

Figure 1: The three-step chain. Steps 1 and 2 are tested under independent training and evaluation noise (Sections 4.1–4.4). The additive substitution of step 3 is tested under independent noise in Section 4.1; the graft onto S4D (right box) was run only under the earlier noise protocol ([Appendix D](#)) and is descriptive. Numbers in the boxes are final-epoch differences in percentage points.

284 4.1. Original transition under independent noise

285 A pre-specified grid of 150 cells (thresholds, labels and writing conse-
 286 quences committed to the public repository before the first cell) re-ran the
 287 central arms with independent training and evaluation noise (Section 3.4);
 288 everything else is unchanged. Each contrast is labelled *holds* if at least two
 289 of the three levels have a positive mean with an interval excluding zero and
 290 none is reversed, *weakened* if exactly one, *does not hold* if none. Relative to
 291 the earlier protocol the arm means moved by -3.4 to $+3.9$ pp in the eight
 292 re-run arms; no contrast changed direction.

Table 3: Independent training and evaluation noise, original XJTU-SY transition ($37.5\text{ Hz}/11\text{ kN} \rightarrow 40\text{ Hz}/10\text{ kN}$). Upper block: macro-F1 (%) at the final epoch, mean over 5 seeds. Lower block: paired differences with 95% t -intervals and the number of seeds (of 5) with a positive difference, and the pre-specified label. [‡]Added during review, not pre-specified.

Arm / contrast	0 dB	-2 dB	-6 dB	label
BM3-frozen	80.3	86.4	87.0	
frozen-all	89.5	91.5	88.9	
S4D	68.3	68.3	68.6	
frozen-gate	70.9	70.7	60.0	
frozen+add	71.3	71.3	60.1	
frozen-all-gate	84.1	82.2	56.3	
frozen-all+add	84.9	83.6	65.6	
frozen-all	+21.3	+23.3	+20.3	
- S4D	[14.6, 27.9] (5/5)	[20.2, 26.4] (5/5)	[17.9, 22.8] (5/5)	holds
BM3-frozen	+12.1	+18.1	+18.4	
- S4D	[0.9, 23.2] (5/5)	[9.8, 26.4] (5/5)	[12.7, 24.0] (5/5)	holds
BM3-frozen	+9.4	+15.7	+27.0	
- frozen-gate	[-0.5, 19.4] (5/5)	[-0.8, 32.1] (4/5)	[7.0, 46.9] (4/5)	weakened
frozen-all	+5.4	+9.3	+32.6	
- frozen-all-gate	[-3.8, 14.6] (3/5)	[-0.5, 19.1] (5/5)	[28.9, 36.3] (5/5)	weakened
frozen-all-gate	+15.9	+14.0	-12.3	
- S4D	[6.9, 24.8] (5/5)	[2.5, 25.4] (4/5)	[-17.5, -7.0] (0/5)	descriptive
BM3-frozen	+9.0	+15.1	+26.8	
- frozen+add	[-0.8, 18.9] (5/5)	[-0.9, 31.0] (4/5)	[6.8, 46.9] (4/5)	descriptive [‡]
frozen-all	+4.6	+7.9	+23.3	
- frozen-all+add	[-4.0, 13.2] (3/5)	[1.0, 14.8] (5/5)	[9.0, 37.5] (5/5)	descriptive [‡]
frozen+add	+0.4	+0.6	+0.1	
- frozen-gate	[-0.6, 1.4] (4/5)	[-0.1, 1.2] (5/5)	[-0.4, 0.7] (3/5)	descriptive
frozen-all+add	+0.8	+1.4	+9.3	
- frozen-all-gate	[-1.5, 3.1] (2/5)	[-1.9, 4.7] (2/5)	[-7.2, 25.8] (5/5)	descriptive

293 *Freezing.* Both freeze contrasts hold. The fully frozen block exceeds S4D
 294 by $+21.3$, $+23.3$, $+20.3$ pp, in every seed at every level; the partially frozen
 295 block exceeds it by $+12.1$, $+18.1$, $+18.4$ pp.

296 *Gate removal.* Removing the gate from the partially frozen block costs
297 +9.4, +15.7, +27.0 pp; the contrast is *weakened* because only the -6 dB
298 interval excludes zero. In the fully frozen block it costs +5.4, +9.3, +32.6 pp,
299 again resolved only at -6 dB. Without its gate the fully frozen block still
300 exceeds S4D by +15.9 and +14.0 pp at 0 and -2 dB (intervals excluding
301 zero) and falls 12.3 pp below it at -6 dB (interval [-17.5, -7.0], all seeds).
302 The gate is therefore not required for a positive advantage over S4D at 0 and
303 -2 dB, but it is required at -6 dB. At the two milder levels the removal cost
304 is +5.4 and +9.3 pp with intervals that include zero, so a contribution of the
305 gate there is neither demonstrated nor excluded. The pre-specified condition
306 for a combined attribution of the fully frozen block's advantage to the gate
307 (gate removal holds and the additive substitution is not rescued) is not met,
308 and we make no such claim.

309 *Multiplicative versus additive combination.* Replacing the gate by an ad-
310 ditive branch of identical size and initial weights cannot be distinguished from
311 removing it in either host (+0.4, +0.6, +0.1 pp and +0.8, +1.4, +9.3 pp; all
312 intervals include zero), and the rescue ratios are small ($\rho = 0.029$, paired-seed
313 interval [0.011, 0.059]; fully frozen host $\rho = 0.194$, [-0.002, 0.326]). Because
314 gate removal is only weakened, the pre-specified rule labels the substitution
315 test *not assessable*. This is a statement that no clear rescue was detected,
316 not that the two arms are equivalent: in the fully frozen host the additive
317 arm is +9.3 pp above the gateless arm at -6 dB (interval [-7.2, 25.8], 5/5
318 seeds). The direct comparison is more informative: the multiplicative gate
319 exceeds the additive replacement by +4.6, +7.9, +23.3 pp in the fully frozen
320 host and by +9.0, +15.1, +26.8 pp in the partially frozen host, with intervals
321 excluding zero at -2 and -6 dB in the former and at -6 dB in the latter
322 (Table 3, descriptive). The ratio ρ is a conditional summary: in 0.5% of the
323 3125 paired seed resamples for the fully frozen host (16.2% for the partially
324 frozen host) at least one noise level was excluded because its denominator
325 fell below 2 pp, as pre-specified; none excluded all three.

326 4.2. Reverse transition under independent noise

327 A second pre-specified grid (60 cells, four arms; thresholds committed
328 before the first cell) repeated the freeze and gate-removal contrasts on the
329 reverse transition, 40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN, with the same bearings in
330 swapped roles, under independent noise. Relative to the earlier protocol the
331 arm means moved by -1.8 to +1.7 pp. This direction is harder for the time-
332 invariant model: on its evaluation set a majority-class predictor scores 46.8%

macro-F1 and a uniformly random one 41.5%, and S4D sits close to those levels.

Table 4: Reverse transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN), independent training and evaluation noise. Macro-F1 (%), mean over 5 seeds, best-epoch / final-epoch; paired differences under final-epoch reporting with 95% t -intervals, seeds positive and label. Fully frozen gate removal and additive substitution were not run in this direction under independent noise (Table 2).

Arm (best / final)	0 dB	-2 dB	-6 dB	label
frozen-all	88.6 / 77.4	88.1 / 77.6	81.1 / 73.3	
BM3-frozen	85.5 / 75.4	86.7 / 78.4	78.5 / 72.2	
frozen-gate	64.0 / 56.1	62.5 / 55.7	55.1 / 50.1	
S4D	61.2 / 49.9	60.2 / 50.9	55.7 / 52.2	
frozen-all	+27.5	+26.7	+21.1	
- S4D	[21.3, 33.8] (5/5)	[19.5, 33.9] (5/5)	[16.2, 26.0] (5/5)	holds
BM3-frozen	+25.5	+27.5	+20.0	
- S4D	[13.9, 37.1] (5/5)	[18.2, 36.9] (5/5)	[8.0, 32.0] (5/5)	holds
BM3-frozen	+19.3	+22.7	+22.0	
- frozen-gate	[2.9, 35.7] (5/5)	[4.1, 41.4] (5/5)	[3.2, 40.9] (5/5)	holds
frozen-gate	+6.2	+4.8	-2.0	
- S4D	[-15.6, 28.0] (2/5)	[-17.5, 27.1] (2/5)	[-19.6, 15.5] (1/5)	descriptive

The fully frozen block exceeds S4D by +27.5, +26.7, +21.1 pp and the partially frozen block by +25.5, +27.5, +20.0 pp; removing the gate from the partially frozen block costs 19.3, 22.7, 22.0 pp, with intervals excluding zero and 5/5 seeds at every level. All three contrasts hold and none is reversed. In this direction gate removal is thus resolved at every level for the partially frozen host, whereas on the original transition it is resolved at -6 dB only; the two directions are reported side by side and not merged. The gateless arm cannot be distinguished from S4D (+6.2, +4.8, -2.0 pp). The contrasts with S4D rest on a baseline close to the trivial floor. The reverse direction shares the eight bearings of the forward one with swapped roles and has a single inner-race bearing in its evaluation set, so it tests direction rather than new bearings.

4.3. Paderborn with shared bearings

On the Paderborn real-damage bearings (1500 $\rightarrow$ 900 rpm), a pre-specified 45-cell grid trained and evaluated on the same four bearings (KA04, KA16, KI04, KI14; different recordings) under independent noise (Table 5). Gate

removal costs +7.6, +6.5, +4.4 pp with intervals excluding zero at all three levels (*holds*); the partially frozen block exceeds S4D by +5.6, +6.5, +8.3 pp and is resolved at −6 dB only (*weakened*). The fully frozen block, the additive arm and the graft were not run in this setting. The companion setting with bearings disjoint from the training set was run only under the earlier protocol, where per-index label agreement between training and evaluation windows is 0.999; it is reported in [Appendix D](#) and is not an independent-noise test. With two bearings per class a bearing’s identity determines its label in the shared setting, so a model can succeed partly by recognising bearing-specific signatures; XJTU-SY, by contrast, evaluates on bearings disjoint from the training ones in each direction.

Table 5: Independent training and evaluation noise, Paderborn shared bearings (KA04, KA16, KI04, KI14; 1500 → 900 rpm). Paired differences under final-epoch reporting with 95% *t*-intervals, seeds positive and pre-specified label. Means: BM3-frozen 81.5 / 83.8 / 85.9, S4D 75.9 / 77.3 / 77.6, frozen−gate 73.8 / 77.3 / 81.5 %.

Contrast	0 dB	−2 dB	−6 dB	label
BM3-frozen	+5.6	+6.5	+8.3	
− S4D	[−4.4, 15.5] (4/5)	[−2.9, 15.9] (4/5)	[2.7, 13.8] (5/5)	weakened
BM3-frozen	+7.6	+6.5	+4.4	
− frozen−gate	[2.7, 12.6] (5/5)	[1.7, 11.3] (4/5)	[3.2, 5.6] (5/5)	holds

4.4. Does the gate’s contribution grow with noise depth?

A contrast that is resolved at one level and not at another does not by itself show that the effect differs between levels. We therefore computed, with seeds paired, the difference between the gate-removal cost at a deeper level and at 0 dB (Table 6). This analysis was added during review and is descriptive.

Only the fully frozen host on the forward transition shows a change whose interval excludes zero at −6 dB (+27.2 pp, 5/5 seeds). The partially frozen host on the forward transition has a positive point estimate with an interval that includes zero; on the reverse transition the change is near zero although the cost itself is large at every level; for Paderborn shared bearings the point estimate is negative. The statement that the gate’s contribution grows with noise depth is therefore supported only by the fully frozen block on the forward transition, and we restrict it to that case throughout; for the other settings the data are compatible with a depth-independent cost.

Table 6: Change of the gate-removal cost (frozen host – its gateless counterpart, percentage points) relative to 0 dB: mean, 95% paired t -interval and seeds (of 5) with a positive change. Independent noise, final epoch. Descriptive; not pre-specified.

Host and transition	−2 vs. 0 dB	−6 vs. 0 dB
	+3.9	+27.2
Forward, fully frozen	[0.5, 7.3] (4/5)	[19.2, 35.2] (5/5)
	+6.2	+17.5
Forward, partially frozen	[−6.3, 18.7] (4/5)	[−1.2, 36.2] (4/5)
	+3.4	+2.7
Reverse, partially frozen	[−1.6, 8.5] (4/5)	[−15.7, 21.1] (2/5)
	−1.1	−3.2
PU shared, partially frozen	[−3.1, 0.9] (1/5)	[−8.4, 1.9] (1/5)

377 4.5. Evidence from the earlier noise protocol

378 Several results were obtained only under the earlier protocol (Table 2,
379 [Appendix D](#)). We list them with the weight we give them; none carries an
380 abstract-level claim by itself.

381 *Graft onto S4D* ([Appendix D.4–Appendix D.5](#)). A gate branch of the
382 original width recovers $R = 0.196$ of the gap between S4D and BM3-
383 frozen; one rebuilt in the donor’s width recovers 0.296 (paired-seed
384 interval 0.14, 0.72), against 0.408 for an additive branch with identical
385 parameters. On the original transition the gate did not recover more
386 than the additive branch; on the reverse transition it recovered some-
387 what more ($\Delta R = 0.153$, paired-seed interval 0.01, 0.34). Descriptive.

388 *Additive substitution on the reverse transition* ([Appendix D.6](#)): not
389 rescued by the pre-specified rule; not re-run under independent noise.

390 *Stem and A removal* ([Appendix D.3](#)): no resolved effect of remov-
391 ing the heavy-tailed A parameterisation; removing the stem improves
392 macro-F1 at 0 and −2 dB (intervals excluding zero) and costs 5.9 pp at
393 −6 dB (interval including zero).

394 *Per-class recall* ([Appendix D.10](#), 20-epoch grid): gate-related differ-
395 ences concentrate on the minority inner-race class.

396 *Paderborn with disjoint bearings* ([Appendix D.8](#)): none of the pre-
397 specified contrasts replicates and no architecture transfers well (47.7–
398 56.9%).

399 *Reporting rule* (Appendix D.9): best-epoch values from the original
 400 grids are optimistic by 0.6–10.6 pp relative to the final epoch; orderings
 401 are unchanged.

402 5. Discussion

403 **What can be credited?** Within this Mamba-3 block, on this testbed
 404 and with noise-aware training, four statements are supported by independent-
 405 noise evidence. (i) Making the scan’s dynamics and read/write coefficients
 406 input-independent did not remove the block’s advantage over S4D on either
 407 transition of one dataset: S4D trails the frozen block by 12–28 pp. This
 408 does not show that input-dependence is useless: it shows that, in this block
 409 and under this budget, learned constants suffice to keep the advantage, and
 410 the two readings (“input-dependence contributes nothing here” and “what it
 411 contributes is recovered by constants”) lead to the same attribution. (ii) The
 412 gate-removal cost is large in the partially frozen block (resolved at -6 dB on
 413 the original transition and at every level on the reverse one) and in the fully
 414 frozen block (resolved at -6 dB). (iii) Without its gate the fully frozen block
 415 still exceeds S4D at 0 and -2 dB, so the gate is not required for a positive
 416 advantage there; the removal cost at those levels ($+5.4$ and $+9.3$ pp) has
 417 intervals that include zero, so a contribution is neither shown nor excluded.
 418 At -6 dB the gateless block falls below S4D. (iv) A multiplicative gate ex-
 419 ceeds an additive branch of identical size and initial weights by $+23.3$ pp at
 420 -6 dB in the fully frozen block and by $+26.8$ pp in the partially frozen block,
 421 whereas the additive arm cannot be distinguished from the gateless one; the
 422 pre-specified rescue test is nevertheless not assessable and equivalence is not
 423 claimed.

424 **Noise-dependent gating.** Only one setting shows a gate contribution
 425 that grows significantly with noise depth: the fully frozen block on the orig-
 426 inal transition (change of the removal cost between 0 and -6 dB $+27.2$ pp,
 427 interval excluding zero; Section 4.4). For the partially frozen block, the re-
 428 verse transition and Paderborn shared bearings the data are compatible with
 429 a depth-independent cost, and the statement in the title is restricted to the
 430 first case. The depth analysis was added during review and is descriptive.

431 **Relation to earlier work.** Representability results for selectivity [8] are
 432 not contradicted by a single ablation on one testbed. The S4D-superiority
 433 finding [4] was obtained on clean classification benchmarks, without matched
 434 noise or a shift of operating condition, so it does not speak to the regime

435 studied here. Two points of contact are nevertheless consistent with it: parity
 436 between S4D and the Mamba-style arms on clean data ([Appendix D](#)), and
 437 the finding that input-dependent selectivity is not needed for the block’s
 438 advantage. What this paper adds is specific to the matched-noise, cross-
 439 condition regime studied here, not a contradiction of the clean-benchmark
 440 result.

441 **Implications for backbone selection and for transplant practice.**
 442 For practitioners choosing backbones for noisy, drifting signal regimes, the
 443 results suggest judging the block as a composition. Whether a gate added
 444 to a different host reproduces its effect was tested only under the earlier
 445 noise protocol ([Appendix D.5](#)): a gate built in the donor’s shape recovered
 446 no more than an additive branch of the same size on the original transition
 447 and somewhat more on the reverse one, and neither design was tuned; we
 448 treat this as descriptive. A hypothesis for future work follows: a recomposed
 449 block guided by the component map (no conv stem, gate retained, S4D-style
 450 *A*) may match or exceed the original on this testbed; we leave it untested,
 451 because post-hoc recombination after seeing the map would not carry the
 452 evidential status of the pre-specified chain.

453 **Why might gating be composition-bound?** We refrain from a mech-
 454 anistic claim beyond the data, but note three compatible observations. First,
 455 the gate-related differences concentrate on the minority, lowest-recall class
 456 ([Appendix D.10](#)), and matched-noise training lifts that class *only when the*
 457 *gate is present*. One hypothesis, which we did not test, is that the gate is the
 458 component through which matched-noise training improves hard-class recall,
 459 i.e., that it exploits noise-aware training rather than acting as a test-time de-
 460 noiser; we measured neither classification margins nor gate activations, and
 461 the reading is in any case bounded by the matched-SNR scope (Section 6).
 462 Second, the multiplicative form admits a per-sample suppression interpre-
 463 tation, but no gate statistics were recorded, so this remains a hypothesis
 464 rather than an observation. Third, the gateless composition falling to or be-
 465 low S4D at the deepest noise levels is compatible with the other components
 466 being calibrated to a gated regime (e.g., the stem may pass features that
 467 are only safe when the gate can veto them); this is the weakest of the three
 468 observations, since the gateless–S4D gap is not resolved with five seeds ([Ap-
 469 pendix D.9](#)). All are testable in future work via pairwise interaction grids
 470 and mismatched-noise protocols.

471 6. Boundaries and Limitations

472 We state the scope plainly.

- 473 1. **Testbed scope.** All claims are bounded to bearing vibration from
474 one pair of operating conditions of one dataset (37.5 Hz/11 kN and
475 40 Hz/10 kN; speed and load both change), binary classification, G-
476 PU execution (no edge/real-time claims). Within each direction the
477 training and evaluation bearings are disjoint; the reverse direction us-
478 es the same eight bearings with their roles swapped, has a single IR
479 bearing in its evaluation set (the forward direction a single IR bearing
480 in its training set) and so tests direction rather than new bearings; the
481 third condition of the dataset has no IR bearing. The seeds do not
482 resample bearings (Section 3.1), so the intervals carry no bearing-level
483 uncertainty. Further independent datasets and multi-class settings are
484 future work.
- 485 2. **Attribution level.** Credit assigned to “the rest of the composition”
486 is by elimination within the enumerated component set; pairwise inter-
487 action terms are not decomposed.
- 488 3. **Transplant results are earlier-protocol and budget-conditional.**
489 The graft onto S4D was run only under the earlier noise protocol, with
490 a matched 50-epoch budget; a longer or graft-tuned budget might re-
491 cover more, and we make no claim beyond *this gate branch, in this*
492 *host, under a matched training budget* (Appendix D.4).
- 493 4. **Matched-SNR scope.** Training and evaluation share the same in-
494 jected SNR throughout (Section 3.1); “robustness” in this paper is
495 matched-noise robustness under noise-aware training, not clean-train/noisy-
496 test generalisation. The gate’s benefit may operate by exploiting matched-
497 noise training (consistent with Appendix D.10) rather than by test-
498 time denoising; attribution under mismatched-noise protocols is untest-
499 ed and may differ.
- 500 5. **Coverage by protocol.** Table 2 states which operations were run
501 under independent noise. Not run under independent noise: the graft
502 onto S4D, the stem and *A* ablations, additive substitution on the re-
503 verse transition, the per-class grid and the disjoint-bearing Paderborn
504 setting. Not run at all: gate removal and additive substitution in the
505 fully frozen block on the reverse transition, and the fully frozen block,
506 additive arm and graft on Paderborn shared bearings.

- 507 6. **Noise-regime coverage.** Under strong coloured noise at -10 dB a
508 TCN baseline slightly overtakes the Mamba-style block, and -10 dB
509 is near-chance for the time-invariant baseline and excluded from ad-
510 judication; the demonstrated advantage regime is matched, Gaussian-
511 dominated noise.
- 512 7. **Conv-stem dual reading.** Its clean-data cost admits both representation-
513 mistransfer and optimisation-difficulty readings, reported in place ([Ap-
514 pendix D.3](#)).
- 515 8. **Reporting rule.** Reported values are final-epoch macro-F1 under
516 a fixed 50-epoch budget. The best-epoch rule of the original grids,
517 for which the early thresholds were pre-specified, is optimistic by 0.6–
518 10.6 pp ([Appendix D.9](#)). The final-epoch re-run of the original-design
519 arms (earlier noise protocol) and the independent-noise re-run cover
520 different sets of cells ([Table 2](#)); levels outside the adjudication band
521 and the per-class grid were not re-run. No source-domain validation
522 rule was available, and because the final epoch of the same evaluation
523 condition has been inspected repeatedly, final-epoch values are not a
524 fresh confirmatory test.
- 525 9. **Scope of the freezing intervention.** The first freeze arm removes
526 the input-dependence of Δ , A , B , C only; the fully input-independent
527 scan is a separate grid that was specified after the earlier grids had
528 been seen. No equivalence margin was pre-specified, so “no measurable
529 cost” means the point estimates favour the frozen arms with intervals
530 including zero, not equivalence. The value stream and the gate in-
531 put remain input-dependent, and freezing is implemented inside this
532 Mamba-3 layer; a conclusion about input-dependence in general would
533 need other blocks and tasks.
- 534 10. **Gate interventions.** Under independent noise, gate removal is re-
535 solved only at -6 dB in the partially and the fully frozen block on the
536 original transition, and without its gate the fully frozen block still ex-
537 ceeds S4D at 0 and -2 dB; the combined statement that the fully frozen
538 block’s advantage is carried by the gate is therefore not made. Where
539 the gate-removal cost is resolved at every level (partially frozen block,
540 reverse transition) it was not tested in the fully frozen block. The ad-
541 ditive substitution could not be distinguished from gate removal, but
542 the multiplicative–additive differences are positive at -6 dB; with gate
543 removal only weakened the pre-specified rescue rule is not assessable,
544 and neither equivalence nor zero rescue is claimed. The rescue ratio

- 545 is a conditional summary (levels with a denominator below 2 pp are
546 excluded in some resamples).
- 547 11. **Noise protocol.** All grids not marked in Table 2 as independent-
548 noise used training and evaluation noise shared by index. Where both
549 protocols exist the arm means differ by at most 4 pp and no contrast
550 changed direction, which is not a guarantee for the arms that were not
551 re-run.
 - 552 12. **Strength of the pre-specification evidence.** Thresholds were pre-
553 specified internally; for the follow-up grids from the width-matched
554 graft onwards the thresholds were also committed to a public repos-
555 itory before the first cell, for the earlier grids the evidence is local
556 file times and run-directory names, not an external time-stamped reg-
557 istry (Section 3.5). The noise-depth analysis (Section 4.4) and the
558 multiplicative-additive differences were added during review and are
559 not pre-specified. The process-level repeat check is a single observa-
560 tion.
 - 561 13. **Construction checks.** For the frozen arm the state-dictionary check
562 compares names and shapes of non-target tensors; value-level compar-
563 ison was applied to the component-ablation and graft arms only.
 - 564 14. **Second dataset.** On Paderborn, the gate-removal cost replicates
565 when training and evaluation share bearings under independent noise,
566 and the partially frozen block’s advantage over S4D is resolved at -6 dB
567 only. With bearings disjoint from the training set (earlier protocol only;
568 per-index label agreement 0.999) nothing replicates and every architec-
569 ture, S4D included, transferred poorly (47.7–56.9% against a balanced-
570 chance level of about 50%). The two Paderborn settings differ in more
571 than bearing overlap, so they do not identify why transfer fails. The
572 fully frozen block was run only in the disjoint setting.
 - 573 15. **Sample size.** Every cell mean uses five seeds. The smallest attainable
574 one-sided Wilcoxon p is 0.031, paired intervals are wide, and several
575 secondary statements (conv stem, A parameterisation, gateless versus
576 S4D) are not resolved at this size. For the original-design grids conclu-
577 sions rest on effect sizes that clear the pre-specified thresholds; for the
578 later grids they rest on paired intervals as defined in Section 3.5. Seeds
579 resample initialisation, batch order and noise (evaluation noise is one
580 bank per seed), not bearings.

581 A single-repeat check of process-level non-determinism (~ 0.7 pp) and the

582 label-hash/adjudication trail are documented in the reproducibility appen-
583 dices ([Appendix A](#)–[Appendix B](#)).

584 7. Conclusion

585 In this Mamba-3 block, on one controlled shift $\times$ noise testbed with
586 matched-noise training and independent training and evaluation noise, mak-
587 ing the scan’s dynamics and read/write coefficients input-independent did
588 not remove the block’s advantage over a time-invariant S4D: the fully frozen
589 block leads by +21.3, +23.3, +20.3 pp on the original transition and by
590 +27.5, +26.7, +21.1 pp on the reverse one. The multiplicative output gate
591 costs most when removed, but this cost is resolved only at -6 dB on the
592 original transition (at every level for the partially frozen block on the re-
593 verse one), and without its gate the fully frozen block still exceeds S4D at 0
594 and -2 dB. Only in the fully frozen block on the original transition does the
595 gate’s contribution grow significantly with noise depth; in the other settings
596 tested the data are compatible with a depth-independent cost. A multiplica-
597 tive gate exceeds an additive branch of identical size at -6 dB in both hosts,
598 although the pre-specified rescue test is not assessable. On Paderborn the
599 gate-removal cost replicates when bearings are shared between training and
600 evaluation. Grafting a gate onto S4D, per-class results, the stem and A ab-
601 lations and the disjoint-bearing Paderborn setting were run only under the
602 earlier noise protocol and are descriptive. The evidence rests on one block,
603 one shift pair per dataset, five seeds and final-epoch reporting under a fixed
604 budget. We offer the chain—freeze, remove, replace—as a protocol for turn-
605 ing architecture comparisons into component-level tests, with two cautions
606 from this study: interventions must be matched in width and capacity, and
607 the noise streams of training and evaluation must be independent.

608 CRediT authorship contribution statement

609 Yi Wang: Conceptualization, Methodology, Software, Validation, Formal
610 analysis, Investigation, Data curation, Writing – original draft, Writing –
611 review & editing, Visualization, Supervision, Project administration.

612 Declaration of competing interest

613 The authors declare that they have no known competing financial inter-
614 ests or personal relationships that could have appeared to influence the work

615 reported in this paper.

616 Declaration of Generative AI and AI-assisted technologies in the 617 writing process

618 During the preparation of this work the authors used LLM-based agents
619 to execute and cross-check pre-specified analysis and adjudication passes as
620 described in Appendix C, and to assist with manuscript drafting and figure
621 generation. After using these tools, the authors reviewed and edited the con-
622 tent as needed and take full responsibility for the content of the published
623 article. All numeric thresholds, adjudication rulings, and scientific interpre-
624 tations reported in this paper were fixed and made by the human authors,
625 as detailed in Appendix C.

626 Data availability

627 The code, pre-specification artifacts, and adjudication trail for this study
628 are available at <https://github.com/jeff-1980/bm3-jizi>. The repository
629 includes all follow-up grids (645 cells with per-epoch curves, including
630 the 210 independent-noise cells of Sections 4.1–4.3), their pre-specifications,
631 drivers, construction checks and decision memos and the harness’s own im-
632 ported modules, so the training entry point runs from a fresh checkout.

633 Appendix A. Fairness protocol

634 *Appendix A.1. Evaluation-set fingerprint*

635 Every reported cell in this study is tagged with `eval_sha256`, a SHA-
636 256 hash of the raw held-out test-label byte array, computed once per run
637 and asserted identical across every arm/condition/seed cell within a grid.
638 It fingerprints *that the same held-out label vector was scored every time*, not
639 model weights or predictions. It is also the split on which the best-epoch rule
640 of Section 3.3 selects epochs. The full value shared by all 865 cells across all
641 five grids is:

642 `6c20b367522ce5db59720120f8a7cd12792c7e070c6ae84bd1985511998727de`

643 This value is recomputed independently at adjudication time (not merely
644 copied from the run log) and cross-checked row-by-row against every grid’s
645 fairness report before any comparison across arms is treated as valid.

646 *Appendix A.2. Per-arm parameter counts*

647 Table A.7 lists every arm’s parameter count, each obtained directly via
 648 `sum(p.numel() for p in model.parameters())` on the instantiated model
 649 (not estimated), cross-verified against an independently maintained archi-
 650 tecture map.

Table A.7: Per-arm parameter counts for every arm of the study (from the `n_params` field recorded in each cell and the harness source).

Arm	Parameters
BM3 (<code>bm3_kin</code> / <code>bm3_nokin</code>)	177,938
BM3-frozen (<code>bm3_frozen</code>)	112,410
S4D (<code>s4d</code>)	100,162
S4D-wide (<code>s4d_wide</code> , capacity control, $1.75\times$)	178,498
frozen-conv (<code>frozen_noconv</code>)	112,026
frozen-gate (<code>frozen_nogate</code>)	79,642
frozen-A (<code>frozen_As4d</code>)	112,410
frozen-trap (<code>frozen_ctrp</code>)	111,906
frozen-rot (<code>frozen_cangle</code>)	104,346
frozen-all (<code>frozen_clti</code>)	103,842
frozen-all-gate (<code>frozen_clti_nogate</code>)	71,074
frozen-all+add (<code>frozen_clti_add</code>)	103,842
frozen+add (<code>bm3_frozen_add</code>)	112,410
S4D+gate (<code>s4d_plus_gate</code>)	116,802
S4D+gate, width-matched (<code>s4d_plus_gate_wm</code>)	198,978
S4D+additive branch (<code>s4d_plus_branch</code>)	198,978

651 *Appendix A.3. Five-batch pre-specification timeline*

652 Table A.8 lists, for each of the five pre-specified grids, the artifact that
 653 fixed its adjudication thresholds, that artifact’s creation/modification date,
 654 and the corresponding grid’s run-directory start time. In every case the
 655 prereg artifact predates the grid it governs.

656 Two honest caveats. First, the baseline batch predates the orchestration
 657 infrastructure used for the other four batches by five days; its prereg trail
 658 rests on the task card’s creation timestamp alone, not on the same automat-
 659 ed executor/reviewer machinery documented in [Appendix C](#), and we state

Table A.8: Pre-specification predates data collection for all five grids.

Grid (cells)	Canonical prereg artifact	Prereg date	Grid start
Baseline (475)	Task-tracking-system card (<code>created:</code> 2026-06-27, mtime 15:02)	2026-06-27	2026-06-27 17:31
Frozen-selectivity (105)	<code>frozensel_prereg.json</code> (m- time 18:26:54)	2026-07-02	2026-07-02 18:27
Component ablation (175)	Task-tracking-system card (<code>created:</code> 2026-07-03), materialised in <code>prereg_provenance.md</code>	2026-07-03	2026-07-03 10:34
Graft (35)	<code>graft_prereg_provenance.md</code> (mtime 21:20:36)	2026-07-05	2026-07-05 22:41
Per-class (75)	Task-tracking-system card (<code>created:</code> 2026-07-06)	2026-07-06	2026-07-07 21:21

660 this rather than implying a uniform process across all five batches. Second,
 661 for the component-ablation grid, two additional JSON files bearing “pre-
 662 reg” in their name exist on disk (`secondary_prereg_20260704-0708` and
 663 `secondary_deltapp_prereg_20260704-0719`) that *post-date* the grid’s start
 664 time; both are explicitly marked void in the corresponding decision artifact’s
 665 `void_preregs` field, and neither was used to adjudicate any result in this pa-
 666 per. They are the on-disk residue of the first post-hoc criterion substitution
 667 described in [Appendix C](#), retained here (rather than deleted) as part of the
 668 released adjudication trail.

669 Appendix B. Confusion matrices and a repeat-run check

670 Appendix B.1. Full confusion-matrix set

671 Table [B.9](#) gives the complete confusion-matrix set underlying the per-
 672 class analysis of [Appendix D.10](#): 5 arms $\times$ 3 conditions (clean, 0 dB, -6 dB),
 673 each summed over 5 seeds (support OR= 17,520, IR= 8,720 per cell). These
 674 counts were dumped per-sample and re-verified against the reported recal-
 675 l/precision/F1 metrics cell-by-cell before use.

Table B.9: Full confusion-matrix set (summed over 5 seeds; support OR=17,520, IR=8,720 per cell). Columns give true→predicted counts.

Arm	Condition	OR→OR	OR→IR	IR→OR	IR→IR
BM3	clean	17,500	20	3,948	4,772
	0 dB	17,395	125	1,364	7,356
	−6 dB	16,613	907	1,664	7,056
BM3-frozen	clean	17,303	217	3,917	4,803
	0 dB	17,089	431	2,060	6,660
	−6 dB	15,634	1,886	2,543	6,177
S4D+gate	clean	17,327	193	4,969	3,751
	0 dB	16,303	1,217	3,322	5,398
	−6 dB	13,267	4,253	3,244	5,476
frozen−gate	clean	16,935	585	5,139	3,581
	0 dB	16,092	1,428	5,406	3,314
	−6 dB	13,200	4,320	5,469	3,251
S4D	clean	16,828	692	3,354	5,366
	0 dB	14,945	2,575	2,897	5,823
	−6 dB	12,477	5,043	3,810	4,910

676 Appendix B.2. Repeat-run check

677 Because two independent process runs of an identical configuration are
678 not bitwise reproducible on this hardware (non-deterministic GPU kernel re-
679 ductions in the selective-scan and normalisation ops), we measured a process-
680 level noise floor directly: the same arm, condition, and seed (`bm3_frozen`,
681 clean, seed 0, 2 epochs) was run twice in separate processes with identical
682 evaluation fingerprints and identical parameter counts confirmed. The two
683 runs’ best macro-F1 differed by

$$|0.69025 - 0.69712| = 0.00687 \approx 0.687 \text{ pp},$$

684 which we report as an observed repeat difference of ~ 0.7 pp (cited in Sec-
685 tion 3.5). It is a single repeat of one arm, one condition and a 2-epoch
686 run, and therefore neither an estimate nor a bound of process variability for
687 full-length runs. It is an order of magnitude below the seed-level variabil-
688 ity observed within the deep-noise band across the curve tables underlying
689 Figures D.2 and D.4 (standard deviations typically 2–9 pp, e.g. BM3-frozen
690 5.7–9.4 pp and S4D 1.7–4.5 pp at $\{0, -2, -6\}$ dB). Arm-level effects in Sec-
691 tion 4 are judged against seed-level variability, not against this single repeat.

692 **Appendix C. Automation disclosure and adjudication gates**

693 *Appendix C.1. Automation disclosure*

694 The mechanical stages of the analysis pipeline—applying pre-specified nu-
695 meric thresholds to aggregated results, re-computing aggregates, re-verifying
696 evaluation fingerprints, and checking each decision against its pre-specified
697 rule—were carried out by LLM-based coding agents under an orchestration
698 harness, with every step auditable in the released trail. Manuscript drafting
699 and figure generation were also assisted by such agents (main text, Declara-
700 tion of Generative AI).

701 *Appendix C.2. Incidents*

702 Three incidents occurred in the original grids, each caught before an er-
703 roneous artefact entered a reported number: (1) on the frozen-selectivity
704 grid an automated decision substituted an unspecified criterion for the pre-
705 specified 10 pp threshold; re-applying the original rule gives the reported out-
706 come, “not the primary mechanism”; (2) on the component-ablation grid an
707 automated pass substituted an interval heuristic for the pre-specified thresh-
708 old, and post-hoc rule files were generated; the true pre-specification predated
709 the grid and was re-applied, and the post-hoc files are retained and marked
710 void; (3) on the graft grid a construction stage was falsely marked as passed;
711 the grid was re-executed from scratch under a corrected task card. Devia-
712 tions in the later follow-up grids (for example a bootstrap made deterministic
713 after first use, and rounding corrections) are recorded in the repository’s de-
714 cision memos. Full descriptions, with dates and artefact names, are given in
715 Supplementary Material S1.

716 *Appendix C.3. Human adjudication gates*

717 All numeric adjudication thresholds, every ruling on an incident, the de-
718 cision to report the incidents, and every interpretive claim in Section 5 were
719 fixed or made by the human authors. No agent could overwrite a protected
720 decision artefact, alter a pre-specified threshold or mark its own output as
721 final.

722 *Appendix C.4. Alignment with Elsevier’s generative-AI policy*

723 In line with Elsevier’s policy, AI tools are not listed as authors and the
724 human authors take full responsibility for the content. We extend the dis-
725 closure beyond the writing process to the adjudication pipeline because that
726 is where automation could affect the reported conclusions.

727 Appendix D. Results under the earlier noise protocol

728 This appendix collects the results obtained while the noise added to win-
729 dow i of the training and the evaluation set was drawn from the same stream
730 (Section 3.4), together with the final-epoch and best-epoch re-runs of the
731 original grids. They are reported in full because they were the pre-specified
732 adjudications of the original design, but they are of lower evidential weight
733 than the independent-noise results of Section 4: no abstract-level claim rests
734 on them alone, thresholds labelled “primary” or “necessary” below refer to
735 the earlier-protocol grid only, and where an independent-noise counterpart
736 exists (Table 2) the counterpart takes precedence. Section 4.5 summarises
737 what these results add.

738 Appendix D.1. Freezing the input-dependent Δ , A , B , C projections (earlier 739 protocol)

Table D.10: BM3, BM3-frozen and S4D, macro-F1 (%), mean over 5 seeds; Δ columns are seed-paired mean differences in percentage points. Upper block: final epoch of the 50-epoch budget on the adjudication band (primary; re-runs of Appendix D.2 and Appendix D.9). Middle block: best epoch on the evaluation condition in the same re-runs (sensitivity). Lower block: best epoch in the original frozen-selectivity grid for levels outside the band (auxiliary; not re-run). Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Condition	BM3	BM3-frozen	S4D	Δ (BM3–frozen)	Δ (frozen–S4D)
<i>Final epoch (primary)</i>					
0 dB	86.9	83.1	67.6	+3.7	+15.5
−2 dB	88.3	87.3	67.5	+0.9	+19.8
−6 dB	83.3	83.1	66.7	+0.2	+16.4
<i>Best epoch (sensitivity, same re-runs)</i>					
0 dB	95.6	90.6	75.9	+5.1	+14.6
−2 dB	95.8	92.3	75.1	+3.5	+17.2
−6 dB	88.6	85.7	68.6	+3.0	+17.1
<i>Auxiliary (best epoch, original grid)</i>					
clean	78.9	78.9	81.0	−0.0	−2.1
−10 dB*	68.1	64.3	52.3	+3.8	+12.0

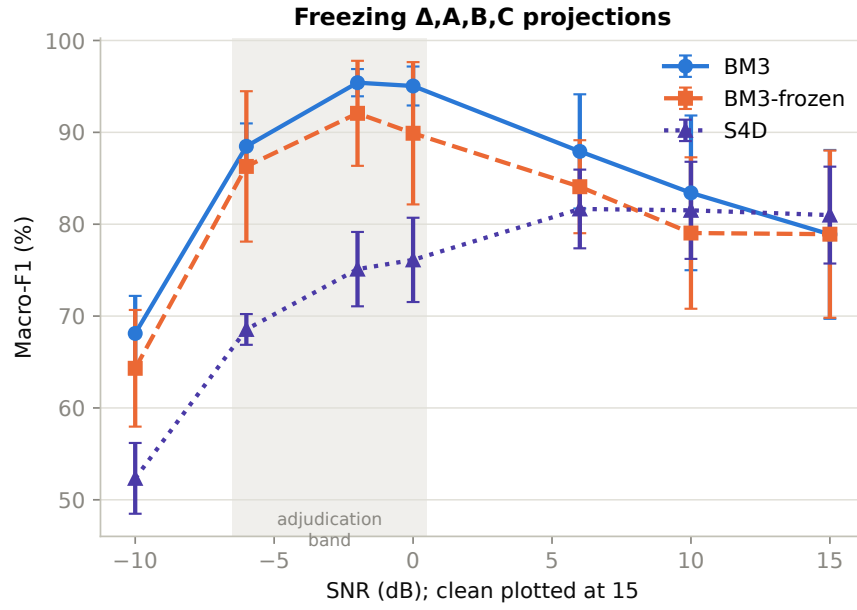


Figure D.2: Macro-F1 vs. SNR for BM3, BM3-frozen, and S4D across the full noise grid (best epoch, original frozen-selectivity grid; final-epoch values on the adjudication band are in Table D.10, and levels outside the band are auxiliary). The frozen arm tracks BM3 closely and stays well above S4D throughout the adjudication band. Earlier noise protocol (training and evaluation noise shared by index).

Freezing the input-dependent Δ , A , B , C projections (Section 3.2) costs at most 3.7 pp anywhere in the adjudication band on the point estimates under final-epoch reporting (+3.7, +0.9, +0.2 pp at 0, -2, -6 dB), while the frozen block retains an advantage of +15.5, +19.8, +16.4 pp over S4D (Table D.10, Figure D.2). Five seeds bound these quantities loosely: paired 95% intervals for $\Delta(\text{BM3-frozen})$ are $[-3.8, 11.3]$, $[-4.5, 6.3]$, $[-12.1, 12.6]$ pp (BM3 ahead in 4/5, 3/5, 2/5 seeds), and for $\Delta(\text{frozen-S4D})$ $[4.3, 26.8]$, $[12.0, 27.7]$, $[4.1, 28.7]$ pp (5/5, 5/5, 4/5 seeds). The pre-specified rule attributed robustness to selectivity only if $\Delta(\text{BM3-frozen}) \geq 10$ pp in deep noise with the frozen arm collapsing toward S4D. It was written for best-epoch reporting, under which the original grid gives $\Delta(\text{BM3-frozen}) = +5.1, +3.3, +2.2$ pp (re-run: +5.1, +3.5, +3.0) and $\Delta(\text{frozen-S4D}) = +13.8, +17.0, +17.7$ pp. Neither condition holds under either rule, so the pre-specified verdict is that these projections are *not the primary mechanism*. This is a failure to meet a confirmation criterion, not a demonstration that freezing is cheap: the intervals above are compatible with differences larger than 10 pp, and no equivalence margin was pre-specified or tested. One limit applies: the frozen arm still contains input-dependent trapezoidal weights and rotation angles, so this result concerns Δ , A , B , C only; Appendix D.2 removes that limit. On clean data (auxiliary, best epoch) S4D matches or slightly exceeds both Mamba-style arms—in line with the clean-benchmark comparison of [4], which, like this auxiliary row, involves no added noise. In the noisy regime, the block’s advantage over S4D is not removed by freezing these four terms, so it must be carried by other parts of the block or by the remaining input-dependent scan terms.

Appendix D.2. Freezing the remaining input-dependent scan terms (earlier protocol)

Table D.11 reports the three layered arms of Section 3.2 on the adjudication band (60 cells: 4 arms $\times$ 3 levels $\times$ 5 seeds $\times$ 50 epochs, the original training code, the same evaluation fingerprint, with the original BM3 arm re-run in the same grid). The question is whether the advantage that survived freezing Δ , A , B , C is carried by the two input-dependent terms the frozen arm retained.

It is not. With the scan’s dynamics and read/write coefficients input-independent, *frozen-all* is +21.9, +24.0 and +21.2 pp above S4D at 0, -2 and -6 dB under final-epoch reporting, positive in 5/5 seeds at every level and with intervals excluding zero—if anything a larger margin than the

Table D.11: Layered freezing of the scan’s input-dependent terms. Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting; Δ columns are paired differences under final-epoch reporting with 95% t -intervals over seeds. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
BM3 (re-run)	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen (Δ, A, B, C)	90.6	83.1	92.3	87.3	85.7	83.1
frozen-trap	95.5	87.6	95.3	89.3	90.5	88.0
frozen-rot	94.6	86.7	94.6	89.1	89.0	86.4
frozen-all	96.5	89.5	96.8	91.5	90.5	87.9
S4D	75.9	67.6	75.1	67.5	68.6	66.7
<i>Paired differences, final-epoch reporting (pp)</i>						
frozen-all − S4D	+21.9 [16.7, 27.1]		+24.0 [21.4, 26.5]		+21.2 [17.5, 24.9]	
BM3-frozen − frozen-all	−6.3 [−17.6, 4.9]		−4.1 [−12.0, 3.7]		−4.8 [−17.8, 8.2]	
BM3 − frozen-all	−2.6 [−8.2, 2.9]		−3.2 [−7.0, 0.6]		−4.5 [−7.2, −1.9]	

777 partially frozen arm’s +15.5, +19.8, +16.4 pp (Table D.19). None of the
778 three freezing steps costs accuracy: the paired differences BM3-frozen −
779 frozen-trap, − frozen-rot and − frozen-all are negative at all three levels
780 (between −1.8 and −6.3 pp under final-epoch reporting), i.e. the more heav-
781 ily frozen arm is numerically ahead, with every interval including zero. The
782 direction is not an artefact of capacity: *frozen-all* has 103,842 parameters
783 against 112,410 for BM3-frozen and 177,938 for BM3. At the deepest level
784 *frozen-all* also exceeds the full BM3 block by 4.5 pp ([−7.2, −1.9] for BM3
785 − frozen-all, 0/5 seeds in BM3’s favour); we report this but do not build on
786 it, since five seeds and one speed pair cannot separate a genuine advantage
787 of a time-invariant scan from a variance effect.

788 Two readings of the whole step are therefore available, and the data do not
789 separate them: input-dependence of the scan contributes nothing measurable
790 here, or it contributes something that the learned constants recover under
791 this training budget. Either way the conclusion for the attribution chain
792 is the same and now rests on a complete intervention rather than a partial
793 one: **the Mamba-style block’s deep-noise advantage over a time-**
794 **invariant S4D does not require any input-dependent term in its**
795 **scan.** What remains to carry it is the rest of the block—the convolutional
796 stem, the heavy-tailed A parameterisation, the normalisations and the output

gate—which is what [Appendix D.3–Appendix D.4](#) examine. The re-run also reproduces the original grids: the BM3 arm’s best-epoch means (95.6, 95.8, 88.6) exceed the original frozen-selectivity grid’s 95.0, 95.4, 88.5 by 0.57, 0.43 and 0.16 pp (a different PyTorch/Triton build, [Appendix D.9](#)).

Appendix D.3. Single-component removal on the frozen base (earlier protocol)

Table D.12: Single-component removal on the frozen base, $\Delta(\text{frozen} - \text{arm})$ in percentage points: final-epoch mean, 95% t -interval over seeds and number of seeds (of 5) with a positive difference (re-runs of [Appendix D.9](#)); “best” is the best-epoch value of the original component-ablation grid, for which the thresholds were pre-specified. Earlier noise protocol (training and evaluation noise shared by index); see [Section 4.1](#) for the independent-noise re-run.

Component removed	0 dB	−2 dB	−6 dB	Adjudication
	+9.1 [2.9, 15.3] (5/5); best +13.1	+15.6 [−0.8, 32.0] (4/5); best +21.1	+22.6 [5.8, 39.4] (5/5); best +23.5	PRIMARY (all $\geq$ 8 pp, both rules)
SiLU gate				
	−13.2 [−24.1, −2.2] (0/5); best −6.6	−7.5 [−14.6, −0.3] (0/5); best −2.7	+5.9 [−4.0, 15.8] (3/5); best +7.9	outside bins; regime trade-off
conv stem ($k=7$)				
	−1.9 [−9.2, 5.3] (2/5); best −2.4	−2.6 [−8.6, 3.4] (2/5); best −2.2	−4.8 [−16.4, 6.9] (2/5); best −4.2	outside bins; inter- vals include 0
heavy-tailed A → S4D-style A				

“Necessary” is used here in the operational sense of this ablation: removing the gate from the frozen block, with the other components and the budget unchanged, removes most of the block’s advantage. The gate result is large, one-signed and grows with noise depth: under final-epoch reporting removal costs 9.1, 15.6, 22.6 pp at 0, −2 and −6 dB (paired 95% intervals [2.9, 15.3], [−0.8, 32.0], [5.8, 39.4]; 5/5, 4/5, 5/5 seeds), above the pre-specified 8 pp threshold at every level; the interval at −2 dB includes zero because of the large seed variance of the gateless arm. Under the best-epoch rule for which the threshold was written, the component-ablation grid gives 13.1, 21.1 and 23.5 pp ([6.5, 19.6], [3.8, 38.5], [8.5, 38.6]; 5/5 seeds at every level), so under the earlier noise protocol the verdict is *primary* under both rules. Under independent training and evaluation noise the same contrast is resolved only at −6 dB ([Section 4.1](#)), so the operational sense of “necessary” holds there

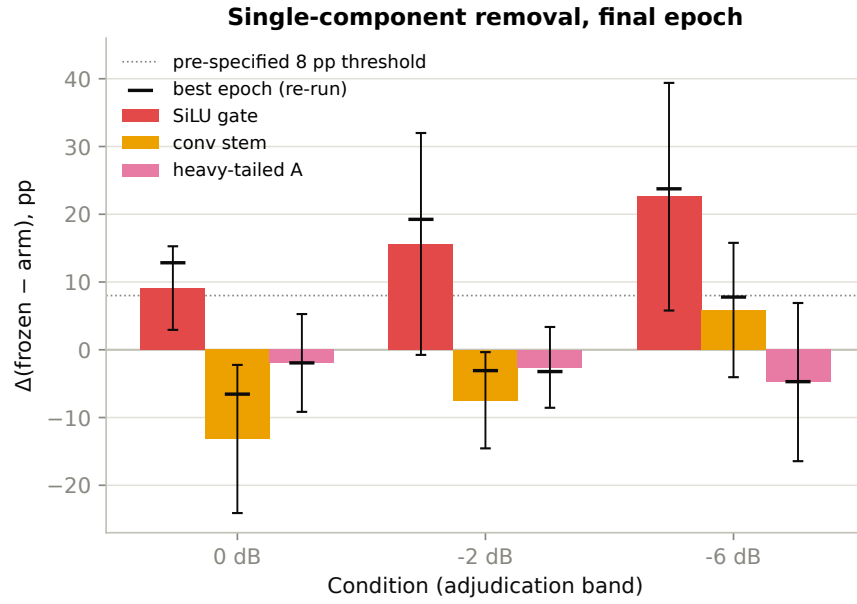


Figure D.3: Component-removal Δ (percentage points, frozen – arm) on the adjudication band for the three single-component ablations on the frozen base: final-epoch means with 95% t -intervals over seeds (bars) and best-epoch means of the same re-runs (black ticks). Only gate removal is large and one-signed at every level and clears the pre-specified 8 pp threshold. Earlier noise protocol (training and evaluation noise shared by index).

816 at the deepest level only. The gateless arm gives up most of the Mamba
 817 advantage but is not resolved against plain S4D: frozen+gate+ S4D is +6.4,
 818 +4.2, -6.2 pp, with every interval including zero. We therefore describe it
 819 as roughly at the level of S4D, below it only on the point estimate at -6 dB;
 820 this suggests, but does not establish, that components interact rather than
 821 add. The other two components tell regime stories rather than robustness
 822 stories. Removing the convolutional stem is beneficial at 0 and -2 dB (frozen
 823 - frozen+conv = -13.2 [-24.1, -2.2] and -7.5 [-14.6, -0.3], with the stem-
 824 less arm ahead in every seed) and costly at -6 dB (+5.9, [-4.0, 15.8], 3/5
 825 seeds); under best-epoch reporting the pattern was the same (-6.6, -2.7,
 826 +7.9 pp). On clean cross-speed data (auxiliary, best epoch, original grid)
 827 the stem costs 21.0 pp (removal yields 98.8 vs. 77.8, with seed-std collapsing
 828 from 8.1 to 0.6 pp)—consistent either with speed-specific local temporal pat-
 829 terns mistransferring across conditions, or with optimisation friction under
 830 the matched budget; we report both readings. Replacing the heavy-tailed
 831 state parameterisation by an S4D-style one is numerically beneficial at all
 832 three levels (by 1.9, 2.6, 4.8 pp under final-epoch and 2.4, 2.2, 4.2 pp under
 833 best-epoch reporting), but every paired interval includes zero, so we draw no
 834 directional conclusion from it.

835 *Appendix D.4. Adding a narrow gate branch to S4D (earlier protocol)*

836 The pre-specified recovery ratio is $R = \text{mean over the band of (S4D+gate-}$
 837 $\text{S4D})/(\text{frozen} - \text{S4D})$. Under final-epoch reporting $R = \mathbf{0.196}$ (per level
 838 0.249, 0.341, -0.003; seed-resampled 95% interval [0.09, 0.31]); under the
 839 best-epoch rule for which it was specified, $R = 0.174$ ([0.04, 0.29]). The pre-
 840 specified bins were $R \geq 0.7$ (confirmed), $0.3 \leq R < 0.7$ (partial recovery) and
 841 $R < 0.3$ (insufficient); both values fall in the last, although the final-epoch in-
 842 terval reaches the 0.3 boundary. The per-level paired gains of S4D+gate over
 843 S4D under final-epoch reporting are +3.9 ([-0.8, 8.6]), +6.8 ([2.4, 11.1]), -0.1
 844 ([-4.0, 3.9]) pp. At the deepest adjudicated level the graft recovers nothing.
 845 Removal and addition are not mirror-image interventions. In BM3 the gate
 846 has its own $64 \rightarrow 128$ projection and multiplies the scan output at the in-
 847 ner width of 128 before the output projection; removing it deletes 32,768
 848 parameters. In the S4D host the grafted gate is a $64 \rightarrow 64$ linear branch
 849 (with bias) computed from the layer input and multiplying the SSM output
 850 before the residual addition; it adds 16,640 parameters over four layers. The
 851 two share the multiplicative SiLU form but not width, projection structure
 852 or parameter count. Removal costs 9–23 pp in the frozen block, whereas this

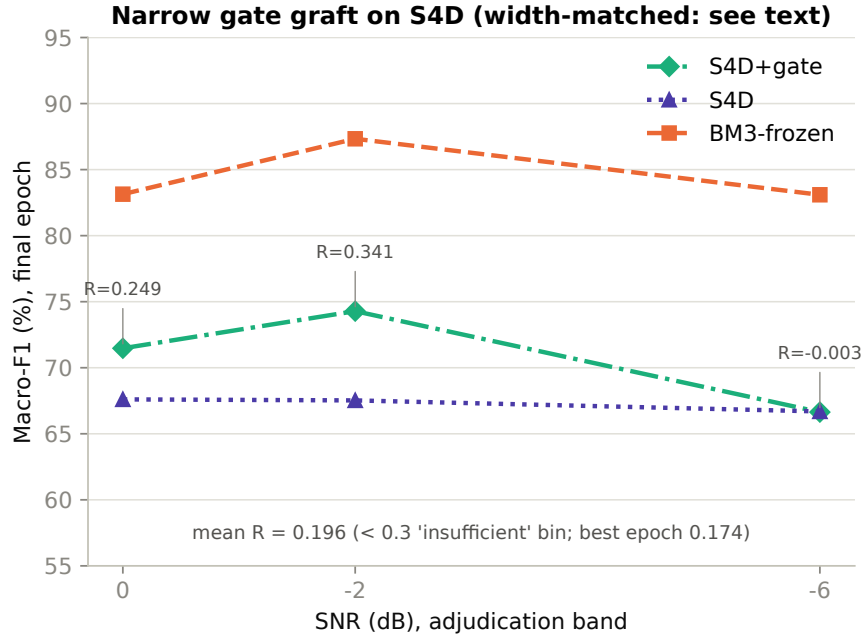


Figure D.4: Graft outcome under final-epoch reporting: S4D+gate, plain S4D, and BM3-frozen on the adjudication band (one re-run grid), with the per-level recovery ratio $R = (\text{S4D+gate} - \text{S4D}) / (\text{frozen} - \text{S4D})$ annotated. Recovery is near zero at the deepest adjudicated noise level. This is the narrow graft; the width-matched graft and its additive control are in Table D.14. Earlier noise protocol (training and evaluation noise shared by index).

Table D.13: Grafting the SiLU gate branch onto plain S4D (S4D unchanged key-by-key; only the gate branch added). Upper block: final epoch, all three arms from the same re-run (Appendix D.9). Lower block: best epoch as pre-specified, with S4D+gate from the graft grid and S4D and BM3-frozen, and hence R , from the component-ablation grid of Appendix D.3 (independent runs of the same arms differ by 0.1–0.7 pp). Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Condition	S4D+gate	S4D	BM3-frozen	Recovery ratio
<i>Final epoch (primary)</i>				
0 dB	71.5	67.6	83.1	0.249
−2 dB	74.3	67.5	87.3	0.341
−6 dB	66.6	66.7	83.1	−0.003
<i>Best epoch (as pre-specified)</i>				
0 dB	79.8	76.2	90.6	0.249
−2 dB	79.4	75.2	92.8	0.239
−6 dB	69.2	68.6	86.0	0.035

853 graft changes S4D by between −0.1 and +6.8 pp (final epoch). What this
854 shows is that *this gate branch, in this host and under this budget, recovers*
855 *little*. Three readings fit the data and cannot be separated here: (i) host de-
856 pendence, i.e. the gate needs the stem or scan context it co-trained with; (ii)
857 implementation differences in width, projection and placement; (iii) trans-
858 plant friction, i.e. the grafted branch is harder to optimise in a foreign host
859 under the matched budget (training converges normally, but convergence is
860 not robustness). Appendix D.5 tests reading (ii) with a width-matched graft
861 and an equal-capacity non-multiplicative control; a tuning or longer-budget
862 comparison selected on a source-domain validation set was not run.

863 The four arms do allow a coarse host-by-gate contrast, $I = [F(\text{frozen}) -$
864 $F(\text{frozen-gate})] - [F(\text{S4D+gate}) - F(\text{S4D})]$, where F is macro-F1. Under
865 last-epoch reporting $I = 5.2$ [2.3, 8.1], 8.9 [−6.2, 23.9] and 22.6 [2.9, 42.4] pp
866 at 0, −2 and −6 dB (seed-matched 95% t -intervals; positive in 5/5, 4/5 and
867 5/5 seeds); under best-epoch reporting in the re-run it is 7.8 [1.4, 14.2], 14.4
868 [−1.4, 30.3] and 23.1 [8.7, 37.4] pp. A positive I means the gate adds more
869 on the frozen host than on S4D. Because the host differs in several respects
870 at once and the two gates are not aligned, I cannot be assigned to a specific
871 component or interaction term, and the seed pairing across arms is nominal

(arms share seed indices, not initialisations).

Appendix D.5. A width-matched graft and an additive control (earlier protocol)

The graft of [Appendix D.4](#) is narrower and smaller than the donor gate. Two further arms rebuild it in the donor’s shape (Table 1). In each layer the layer input is projected to x and z of width 128 (the frozen block’s inner width), the S4D layer runs on x at width 128 with state size 64—so that its own parameter count and total state size equal the host’s 64-wide, state-128 layer—and an output projection maps back to width 64. *S4D+gate* (*wm*) combines $y \odot \text{SiLU}(z)$, as the donor does; *S4D+branch* uses the same modules, parameters and initial weights but combines $y + \text{SiLU}(z)$. Both have 198,978 parameters, more than BM3-frozen (112,410) and BM3 (177,938), which biases the test towards recovery. A construction check with forward hooks confirms that the tensor entering the output projection equals the intended formula for each arm and not the other, and that the two arms are identical at initialisation. The thresholds were committed publicly before the grid ran (Section 3.5): R as before, under final-epoch reporting; the verdict is INCONCLUSIVE whenever the seed-resampled interval crosses a bin boundary, in which case we write at the tier less favourable to the claim; and a gate-specific reading requires $\Delta R = R_{\text{gate}} - R_{\text{branch}} > 0.1$ with an interval excluding zero.

Table D.14: Width-matched graft and additive control on the original transition. Macro-F1 (%), mean over 5 seeds, final-epoch reporting; R is the mean over the three levels of (arm – S4D)/(BM3-frozen – S4D), with a seed-resampled 95% interval. S4D, BM3-frozen and the narrow graft are the re-runs of [Appendix D.9](#). Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm	0 dB	−2 dB	−6 dB	R	95% interval
S4D	67.6	67.5	66.7	–	–
S4D+gate (narrow)	71.5	74.3	66.6	0.196	[0.09, 0.31]
S4D+gate (wm)	70.1	73.4	73.8	0.296	[0.16, 0.52]
S4D+branch	73.1	76.2	73.8	0.408	[0.31, 0.60]
BM3-frozen	83.1	87.3	83.1	1	–
$\Delta R = \text{S4D+gate (wm)} - \text{S4D+branch}$				−0.112	[−0.26, 0.02]

The width-matched graft recovers $R = 0.296$ of the gap (per level 0.159, 0.298, 0.431). Its interval, [0.16, 0.52] under the pre-specified bootstrap

895 ([0.14, 0.72] under paired-seed resampling), crosses the 0.3 boundary, so the
 896 pre-specified verdict is INCONCLUSIVE between “insufficient” and “par-
 897 tial”, and we report it at the less favourable tier: a graft built in the donor’s
 898 shape recovers part of the gap. The recovery is not gate-specific. The addi-
 899 tive branch, with the same parameters, recovers at least as much ($R = 0.408$),
 900 $\Delta R = -0.112$ $[-0.26, 0.02]$ (paired-seed $[-0.20, 0.06]$), and at 0 dB the gat-
 901 ed arm is below the additive one by 3.0 pp ($[-4.5, -1.5]$, 0/5 seeds). With
 902 frozen—all in the denominator the gated arm’s ratio is 0.231 $[0.13, 0.33]$; un-
 903 der best-epoch reporting the two ratios are 0.459 and 0.483. What a graft
 904 recovers on this transition therefore follows the added structure—a wider
 905 channel axis and input/output projections—rather than the multiplicative
 906 interaction. Reading (ii) of [Appendix D.4](#) is partly confirmed, in that the
 907 narrow graft understated what added structure can recover; but in the S4D
 908 host the gate did not contribute measurably beyond an equal-capacity ad-
 909 ditive branch, whereas removing it from the frozen Mamba-3 block costs
 910 9–23 pp under the same rule (Table [D.19](#)).

911 *Appendix D.6. Replacing the gate in its native host (earlier protocol)*

912 The previous subsection changes the host and keeps the operator. A pre-
 913 specified grid (15 cells, original transition) does the converse: it keeps the host
 914 and changes the operator. In *frozen+add* the scan kernel of every BM3-frozen
 915 layer is called without gating and the layer output becomes $W_{\text{out}}(y + \text{SiLU}(z))$;
 916 modules, parameter count (112,410) and initial weights are bitwise those of
 917 BM3-frozen, and a construction check confirms that the donor operator is
 918 $y \odot \text{SiLU}(z)$ and that the new arm computes $y + \text{SiLU}(z)$. The pre-specified
 919 statistic is the native rescue ratio $\rho = (\text{add} - \text{no gate})/(\text{frozen} - \text{no gate})$,
 920 mean over levels, final epoch, with a seed-resampled interval; *not rescued*
 921 requires $\rho < 0.3$ with an upper bound below 0.5.

922 The additive branch does not measurably rescue the loss. *frozen+add*
 923 cannot be distinguished from the gateless arm at any level ($+0.6$, -0.2 ,
 924 $+0.4$ pp, every interval including zero), and the multiplicative arm exceeds
 925 it by 8.5, 15.8, 22.2 pp. The rescue ratio is $\rho = 0.024$ (per level 0.070,
 926 -0.015 , 0.016), with a 95% interval of $[-1.159, 0.486]$ under the pre-specified
 927 bootstrap, which resampled each arm and level independently; resampling
 928 seeds jointly across arms and levels, as the paired design warrants, gives
 929 $[-0.043, 0.136]$. Under best-epoch reporting $\rho = 0.005$. By the pre-specified
 930 rule the verdict is *not rescued*: the point-estimate rescue is small and below

Table D.15: Gate replaced by an additive branch inside BM3-frozen, original and reverse transitions. Macro-F1 (%), mean over 5 seeds; paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

	0 dB		−2 dB		−6 dB	
Arm	best	final	best	final	best	final
<i>Original transition (37.5 Hz/11 kN → 40 Hz/10 kN)</i>						
BM3-frozen ($y \odot \text{SiLU}(z)$)	90.6	83.1	92.3	87.3	85.7	83.1
frozen+add ($y + \text{SiLU}(z)$)	78.0	74.7	72.6	71.5	62.3	60.9
frozen−gate (no z)	77.7	74.0	73.1	71.7	61.9	60.5
	+8.5		+15.8		+22.2	
BM3-frozen − frozen+add	[1.1, 15.9] (5/5)		[−1.2, 32.9] (4/5)		[4.6, 39.9] (5/5)	
	+0.6		−0.2		+0.4	
frozen+add − frozen−gate	[−1.7, 3.0] (3/5)		[−3.4, 3.0] (3/5)		[−0.7, 1.4] (3/5)	
<i>Reverse transition (40 Hz/10 kN → 37.5 Hz/11 kN)</i>						
BM3-frozen ($y \odot \text{SiLU}(z)$)	81.1	74.7	84.0	76.7	80.8	73.9
frozen+add ($y + \text{SiLU}(z)$)	64.3	56.5	62.2	54.4	55.7	50.5
frozen−gate (no z)	63.8	56.2	62.0	54.7	57.1	50.0
	+18.1		+22.3		+23.4	
BM3-frozen − frozen+add	[1.2, 35.0] (5/5)		[3.5, 41.0] (5/5)		[6.8, 40.0] (5/5)	
	+0.3		−0.3		+0.6	
frozen+add − frozen−gate	[−0.6, 1.2] (3/5)		[−1.0, 0.4] (2/5)		[−0.2, 1.4] (5/5)	

931 the pre-specified criterion, which is not a demonstration that the two arms
932 are equivalent.

933 A further pre-specified grid (15 cells) repeats the substitution on the
934 reverse transition (Appendix D.7), with BM3-frozen and frozen–gate taken
935 from the reverse grid. The outcome is the same (Table D.15, lower block):
936 frozen+add cannot be distinguished from the gateless arm at any level (+0.3,
937 −0.3, +0.6 pp, every interval including zero), and the multiplicative arm
938 exceeds it by +18.1, +22.3, +23.4 pp (intervals excluding zero, 5/5 seeds at
939 every level). The rescue ratio is $\rho = 0.009$ (per level 0.017, −0.014, 0.023),
940 with a 95% interval of $[-0.943, 0.362]$ under the pre-specified bootstrap and
941 $[0.002, 0.018]$ under paired-seed resampling; under best-epoch reporting $\rho =$
942 −0.005. The pre-specified verdict is again *not rescued*.

943 Together with Appendix D.5 this gives the two halves of a symmetric
944 test. In the S4D host the gate did not recover more than an additive branch of
945 the same size on the original transition and recovered somewhat more on the
946 reverse one; in the Mamba-3 host an additive branch of the same size could
947 not be distinguished from removing the gate, on both transitions. Under the
948 earlier noise protocol this pattern is specific to the multiplicative operator
949 in its native host; Section 4.1 shows that the size of the gate’s contribution
950 there depends on noise depth.

951 *Appendix D.7. The reverse transition (earlier protocol)*

952 A pre-specified grid repeats the chain’s central contrasts on the reverse
953 transition, 40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN (75 cells: frozen–all, BM3-frozen,
954 frozen–gate, S4D and S4D+gate (wm), at three levels and five seeds; Sec-
955 tion 3.1 for the data). The independent-noise re-run of four of these arms
956 is Section 4.2. This direction is harder for the time-invariant model: on
957 its evaluation set, predicting the majority class gives a macro-F1 of 46.8%
958 and uniform random guessing 41.5%, and S4D sits close to those levels (Ta-
959 ble D.16).

960 The first two steps replicate. The fully input-independent scan exceeds
961 S4D by +26.3, +25.6 and +22.2 pp at 0, −2 and −6 dB, with intervals ex-
962 cluding zero and 5/5 seeds at every level, so by the pre-specified rule (at
963 least two of three levels) the first step is replicated on the reverse transition.
964 Removing the gate from BM3-frozen costs 18.4, 22.0 and 24.0 pp, above the
965 8 pp threshold at all three levels, with intervals excluding zero and 5/5 seeds,
966 and the gateless arm is indistinguishable from S4D (intervals include zero).
967 On the third step, which the pre-specification treats as descriptive here, the

Table D.16: Reverse transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN). Macro-F1 (%), mean over 5 seeds, best-epoch and final-epoch reporting; paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm (best / final)	0 dB	−2 dB	−6 dB
frozen—all	86.3 / 77.5	87.8 / 77.9	81.7 / 74.0
BM3-frozen	81.1 / 74.7	84.0 / 76.7	80.8 / 73.9
S4D+gate (wm)	79.0 / 72.9	80.5 / 73.9	78.5 / 72.2
frozen—gate	63.8 / 56.2	62.0 / 54.7	57.1 / 50.0
S4D	61.2 / 51.2	61.0 / 52.4	54.6 / 51.8
	+26.3	+25.6	+22.2
frozen—all − S4D	[21.5, 31.0] (5/5)	[21.1, 30.1] (5/5)	[18.8, 25.6] (5/5)
	+18.4	+22.0	+24.0
BM3-frozen − frozen—gate	[1.2, 35.6] (5/5)	[3.4, 40.6] (5/5)	[7.6, 40.3] (5/5)
	+5.0	+2.4	−1.8
frozen—gate − S4D	[−17.6, 27.7] (2/5)	[−19.2, 23.9] (2/5)	[−18.9, 15.3] (1/5)
	+21.7	+21.5	+20.5
S4D+gate (wm) − S4D	[15.4, 28.0] (5/5)	[17.0, 26.0] (5/5)	[15.7, 25.2] (5/5)

width-matched graft recovers most of the gap, $R = 0.911$ [0.82, 1.04] (0.863 with frozen—all in the denominator). A further pre-specified grid (15 cells) adds the same-size additive control in this direction: it reaches 67.8, 72.1, 68.4% (final epoch) and recovers $R = 0.758$ [0.66, 0.88]. The gate arm exceeds it by +5.1, +1.7, +3.8 pp (only the −6 dB interval excludes zero), and the difference in recovery is $\Delta R = 0.153$ [0.06, 0.26]; under best-epoch reporting it shrinks to 0.037 [−0.1, 0.17]. By the pre-specified descriptive labels this is *unresolved*: in this direction the gate carries a small increment over the additive branch, whereas on the original transition it carried none ($\Delta R = -0.112$). Most of the reverse-direction recovery is therefore available to an additive branch of the same size, and the remaining gate increment is direction-dependent. Both ratios rest on an S4D baseline close to the trivial floor (51–52% against 46.8% for the majority class). Both statements carry the limits of this pair: the same eight bearings with their roles swapped, and a single inner-race bearing in the evaluation set.

Appendix D.8. Paderborn: disjoint and shared bearings (earlier protocol)

A pre-specified grid (75 cells; thresholds, a ceiling-and-floor check and the writing consequences committed before the first cell) repeats the three central

986 contrasts on the Paderborn KAt data [16], using real-damage bearings only.
987 Models train on two outer-race (KA04, KA16) and two inner-race bearings
988 (KI04, KI14) at 1500 rpm and are evaluated on five different bearings (KA15,
989 KA22; KI17, KI18, KI21) at 900 rpm, with load and radial force unchanged;
990 the vibration channel is resampled to 25.6 kHz so that a window spans 80 ms
991 as in XJTU-SY, and everything else (noise, training, arms, budget) is as in
992 the main grids. The pre-specified smoke check (two arms, one extra seed)
993 passed its three criteria, the learnability criterion by 0.3 pp. A second pre-
994 specified grid (45 cells, three arms) then used the same four bearings for
995 training and evaluation, at the same two speeds, from different recordings.
996 This changes the evaluation bearings, their damage profiles, the number of
997 windows and the class balance together, so the two settings are two protocols
998 rather than a single-factor manipulation.

Table D.17: Paderborn, real-damage bearings, 1500 rpm $\rightarrow$ 900 rpm. *Disjoint*: training on KA04, KA16, KI04, KI14 and evaluation on KA15, KA22, KI17, KI18, KI21. *Shared*: training and evaluation on KA04, KA16, KI04, KI14 (different recordings). Upper block: macro-F1 (%) at the final epoch, mean over 5 seeds (majority-class macro-F1: 37.5% disjoint, 33.3% shared). Lower block: paired differences under final-epoch reporting, 95% t -interval and seeds (of 5) with a positive difference. “-”: arm not run in that setting. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm / contrast	Disjoint bearings			Shared bearings		
	0 dB	-2 dB	-6 dB	0 dB	-2 dB	-6 dB
BM3-frozen	51.9	50.3	47.7	84.5	85.8	86.6
frozen-all	54.2	51.3	47.8	-	-	-
S4D	55.6	53.9	51.1	76.1	77.3	77.9
frozen-gate	56.9	54.9	51.1	75.5	78.1	81.3
frozen+add	56.9	54.8	51.0	-	-	-
	-3.8	-3.6	-3.4	+8.4	+8.5	+8.7
BM3-frozen	[-11.9, 4.3]	[-12.6, 5.4]	[-8.9, 2.1]	[-1.3, 18.1]	[0.2, 16.8]	[4.3, 13.0]
- S4D	(1/5)	(2/5)	(1/5)	(4/5)	(5/5)	(5/5)
	-5.0	-4.6	-3.3	+9.0	+7.7	+5.3
BM3-frozen	[-8.8, -1.3]	[-9.2, 0.1]	[-7.1, 0.4]	[5.2, 12.8]	[5.0, 10.4]	[2.4, 8.2]
- frozen-gate	(0/5)	(0/5)	(0/5)	(5/5)	(5/5)	(5/5)
	-1.4	-2.5	-3.3			
frozen-all	[-4.0, 1.2]	[-5.3, 0.2]	[-5.8, -0.8]			
- S4D	(1/5)	(1/5)	(0/5)	-	-	-
	0.0	0.0	-0.1			
frozen+add	[-0.9, 0.8]	[-0.4, 0.3]	[-0.4, 0.2]			
- frozen-gate	(2/5)	(3/5)	(1/5)	-	-	-

999 None of the pre-specified contrasts replicates (Table D.17). The fully

input-independent block does not exceed S4D (frozen-all – S4D: -1.4 , -2.5 , -3.3 pp, the interval at -6 dB excluding zero in the negative direction), nor does BM3-frozen (-3.8 , -3.6 , -3.4 pp). Removing the gate does not cost accuracy: BM3-frozen – frozen-gate is -5.0 , -4.6 , -3.3 pp, below zero in every seed at every level. With no gate effect to rescue, the native-substitution test is not assessable; the additive and gateless arms again cannot be distinguished (0.0 , 0.0 , -0.1 pp). The pre-specified labels are therefore *not replicated* for both freeze contrasts and for gate removal, and *not assessable* for the substitution test.

Two observations bound how this result can be read, and neither changes the labels. First, the testbed discriminates poorly: every arm ends between 47.7 and 56.9% macro-F1 at every level (majority-class predictor 37.5% , a balanced random predictor about 50%), all arms fit the training condition (the gated arms reach the lowest training loss), and none transfers far above the level reached after the first epoch. Second, where differences are resolved they run against the Mamba-3 arms, which reach lower training loss and transfer slightly worse. In the disjoint setting, therefore, the block’s advantage under matched noise and drift is not observed; we do not know whether this reflects a property of the PU transition (speed reduction to 900 rpm with disjoint bearings), the protocol transferred unchanged from XJTU-SY, or a limit of the XJTU-SY result itself, which the shared-bearing setting below separates.

Shared bearings. With bearing identity shared between training and evaluation, the same condition pair becomes transferable: every arm ends between 75.5 and 86.6% at every level, and the pre-specified transferability gate (5-seed mean $\geq 65\%$ with one-sided 95% lower bound $> 60\%$ at two of three levels for at least one arm) is passed by all three arms at all three levels (smallest lower bound 71.3%). Within this setting both effects replicate under the pre-specified rule (Table D.17, right). BM3-frozen exceeds S4D by $+8.4$, $+8.5$, $+8.7$ pp at 0 , -2 , -6 dB, with intervals excluding zero at -2 and -6 dB; removing the gate costs $+9.0$, $+7.7$, $+5.3$ pp, with intervals excluding zero at all three levels and in every seed; no level runs in the opposite direction. The gateless arm returns to the level of S4D (frozen-gate – S4D: -0.6 , $+0.8$, $+3.4$ pp, all intervals including zero), as on XJTU-SY. Under best-epoch reporting the freeze contrast is resolved at all three levels ($+6.9$, $+10.8$, $+13.7$ pp) and gate removal only at -6 dB ($+1.5$, $+3.7$, $+6.4$ pp). The cost of gate removal reaches the 8 pp threshold used on XJTU-SY at one level only; we compare directions, not magnitudes, across datasets.

Read together, the two protocols give different results: with shared bearings every arm, S4D included, ends 18.6–38.8 pp higher than with disjoint bearings (independent seeds; per-arm Welch intervals all exclude zero), and the block’s advantage and the cost of gate removal reappear. Because the evaluation bearings, class balance and sample size change together, this does not isolate bearing overlap as the cause. With two bearings per class in the shared setting, a bearing’s identity also determines its label, so a model can succeed partly by recognising bearing-specific signatures. XJTU-SY differs from both Paderborn settings: its training and evaluation bearings are disjoint in each direction, and the block transfers there. The shared-setting contrasts were re-run under independent noise (Section 4.1); *frozen—all* was not run on Paderborn.

Appendix D.9. Sensitivity to the reporting rule (earlier protocol)

The main tables use the final epoch of the 50-epoch budget (Section 3.3). The original grids recorded instead the best epoch on the evaluation condition, and their pre-specified thresholds were written for that rule. To measure how much the rule shapes the conclusions, every arm of the chain was re-run over the adjudication band with per-epoch evaluation, using the original training code and the same evaluation fingerprint: the four arms BM3-frozen, S4D, frozen–gate and S4D+gate (60 cells), the BM3 arm alongside the layered freezes (Appendix D.2), and the capacity control and the two other single-component removals, S4D-wide, frozen–conv and frozen–A (45 cells, pre-specified with a rule for changes of sign). The re-runs used a different PyTorch/Triton build from the original grids; their best-epoch values reproduce the original arm-by-condition means to within -0.6 to $+1.4$ pp across the eight re-run arms (the largest deviation is in the high-variance frozen–gate arm at -2 dB).

Best-epoch values exceed final-epoch values by 0.6–10.6 pp across the 24 arm-by-level means, so best-epoch absolute levels are optimistic. The reduction is of similar size for the arms being contrasted (e.g., BM3-frozen -7.4 , -5.0 , -2.6 pp and S4D -8.3 , -7.6 , -1.9 pp at 0, -2 , -6 dB), and the contrasts that carry the argument are correspondingly stable (Table D.19).

Selectivity. The frozen block’s advantage over S4D is $+15.5$, $+19.8$ and $+16.4$ pp (and $+21.9$, $+24.0$, $+21.2$ pp for the fully frozen scan, Appendix D.2), versus $+14.6$, $+17.2$ and $+17.1$ pp under best-epoch reporting in the re-run. It remains positive in 5/5, 5/5 and 4/5 seeds, with intervals excluding zero at every level. *Gate.* Removing the gate costs 9.1, 15.6 and 22.6 pp

Table D.18: Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting in the re-runs, for every arm of the chain on the adjudication band. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
BM3	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen	90.6	83.1	92.3	87.3	85.7	83.1
S4D	75.9	67.6	75.1	67.5	68.6	66.7
S4D-wide	77.8	67.3	73.3	66.5	69.0	67.1
frozen−gate	77.7	74.0	73.1	71.7	61.9	60.5
frozen−conv	97.1	96.3	95.4	94.8	77.9	77.2
frozen−A	92.5	85.1	95.6	89.9	90.4	87.9
S4D+gate	80.9	71.5	79.9	74.3	69.2	66.6

Table D.19: Paired differences under final-epoch reporting, in percentage points: mean, 95% t -interval over seeds, and number of seeds (of 5) with a positive difference. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Contrast	0 dB	−2 dB	−6 dB
	+15.5	+19.8	+16.4
BM3-frozen − S4D	[4.3, 26.8] (5/5)	[12.0, 27.7] (5/5)	[4.1, 28.7] (4/5)
	+9.1	+15.6	+22.6
BM3-frozen − frozen−gate	[2.9, 15.3] (5/5)	[−0.8, 32.0] (4/5)	[5.8, 39.4] (5/5)
	+6.4	+4.2	−6.2
frozen−gate − S4D	[−2.9, 15.8] (4/5)	[−14.5, 22.9] (3/5)	[−23.5, 11.2] (1/5)
	+3.9	+6.8	−0.1
S4D+gate − S4D	[−0.8, 8.6] (4/5)	[2.4, 11.1] (5/5)	[−4.0, 3.9] (2/5)

1075 (12.8 to 23.8 pp under best-epoch reporting in the re-run), above the 8 pp
 1076 pre-specified threshold at every level; the interval at -2 dB includes zero
 1077 ($[-0.8, 32.0]$ pp, 4/5 seeds) because of the large seed variance of the gateless
 1078 arm (standard deviation 14.6 pp at -2 dB). *Gateless versus S4D*. Under final-
 1079 epoch reporting the gateless arm is not below S4D at 0 and -2 dB (point
 1080 estimates $+6.4$ and $+4.2$ pp) and is below it at -6 dB (-6.2 pp); all three
 1081 intervals include zero. The statement that gate removal “falls below S4D”
 1082 is therefore supported only as a point-estimate observation at the deepest
 1083 level, and we phrase it as “to or below” elsewhere. *Graft*. The recovery
 1084 ratio is 0.249, 0.341 and -0.003 per level and $R = 0.196$ on average (best-
 1085 epoch re-run: 0.220), again in the pre-specified “insufficient” bin on the point
 1086 estimate. The seed-resampled 95% interval is $[0.09, 0.31]$ and so reaches the
 1087 0.3 boundary: with five seeds the data support “far below confirmation (0.7)
 1088 and probably below 0.3”, not a precise value of R . The graft gain over S4D
 1089 is resolved at -2 dB only ($+6.8$ pp, $[2.4, 11.1]$) and is indistinguishable from
 1090 zero at -6 dB.

1091 *Remaining arms*. By the rule pre-specified for the 45-cell grid, no arm
 1092 changes sign between the two rules at any level where both values exceed
 1093 1 pp, so all main-table entries are reported under the final-epoch rule with
 1094 unchanged wording. The capacity control holds: BM3-frozen exceeds S4D-
 1095 wide by $+15.9$, $+20.8$, $+16.0$ pp (intervals excluding zero at all three levels),
 1096 and S4D-wide $-$ S4D is -0.4 , -1.0 , $+0.4$ pp (intervals including zero). For
 1097 the stem the final-epoch rule sharpens the regime trade-off (intervals at 0
 1098 and -2 dB now exclude zero) without reversing it.

1099 Because the final epoch of the same evaluation condition has been in-
 1100 spected repeatedly during this study, last-epoch values are also not a fresh
 1101 confirmatory test. The reporting rule therefore changes absolute levels and
 1102 the strength of the gateless-versus-S4D statement, but not the ordering BM3-
 1103 frozen $>$ {S4D+gate, frozen $-$ gate, S4D}, the size of the frozen block’s ad-
 1104 vantage over S4D, or the verdicts of the pre-specified gate and graft rules.

1105 *Appendix D.10. Per-class recall (earlier protocol)*

1106 Per-class analysis (a separate 75-cell grid at 20 epochs with best-epoch
 1107 selection on the evaluation condition: 5 arms $\times$ {clean, 0, -6 dB} $\times$ 5 seeds,
 1108 per-sample predictions dumped and re-verified against reported metrics cell-
 1109 by-cell): the inner-race class is hard from the outset (clean recall 0.41–0.62
 1110 vs. 0.96–1.00 for outer-race). Its trajectory under noise is governed by gate
 1111 presence: in all three gated arms IR recall at -6 dB exceeds its clean value

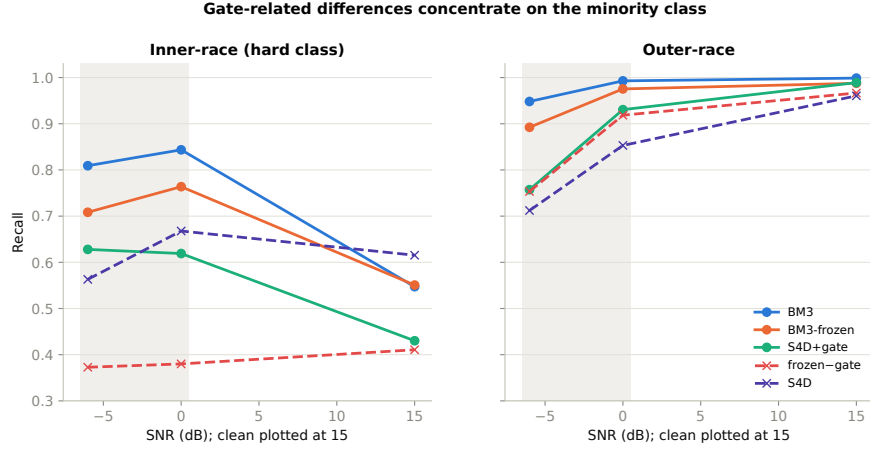


Figure D.5: Per-class recall (inner-race, IR, and outer-race, OR) as a function of noise depth, split by gate presence. Gated arms lift IR recall under matched noise; gateless arms let it deteriorate. Earlier noise protocol (training and evaluation noise shared by index).

(e.g., $0.55 \rightarrow 0.81$ for BM3), while in both gateless arms it is below its clean value (frozen-gate: $0.41 \rightarrow 0.37$; S4D: $0.62 \rightarrow 0.56$, non-monotone, rising to 0.67 at 0 dB) (Figure D.5). Both ablation directions hit IR hardest—gate removal costs IR $2.3\text{--}3.8\times$ more F1 than OR at every level (F1 from predictions pooled over seeds), and the graft’s noise-time gains also concentrate on IR (-6 dB: $+6.8$ pp IR vs. $+4.2$ pp OR). Because macro-F1 averages the two classes, the arm-level robustness gaps of Appendix D.1–Appendix D.4 are carried substantially by the hard class. The gate-related differences are therefore class-concentrated rather than uniform, and fall on the class with the lowest recall. We did not measure classification margins, so we do not describe IR as a “low-margin” class. Because this grid is separate from the main 50-epoch grid and the IR class comes from a single training bearing (Section 3.1), these results are descriptive and do not decompose the main-grid effects.

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